Possible Futures in Sight: Disability Service Facilities and Local Fertility*

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Abstract

This paper examines whether fertility decisions respond to the perceived risk of having a child with a disability and associated care needs. Survey evidence suggests that these concerns are often as important as work-related concerns for childbearing intentions. To examine whether these concerns also shape observed fertility, I study areas around disability service facilities, where incidental encounters with users and caregivers may make disability and care needs more salient. Using spatially granular Japanese census data, I find lower child-to-woman ratios near these facilities, equivalent to approximately 1–1.8% of Japan's births. Consistent with this mechanism, child-to-woman ratios are also lower along routes between facilities and railway stations, and both associations weaken with car dependence. Placebo analyses and adjustments for migration, land prices, and safety provide evidence against alternative explanations. These findings suggest that perceived disability risks or anticipated care needs may discourage childbearing.

Keywords: fertility decisions; disability; risk perceptions; incidental social contact; neighborhood effects

JEL codes: I10, J13, I14, R23, Z13

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1 Introduction

Amid the global decline in fertility, understanding what shapes decisions about whether and when to have children is increasingly important (Doepke et al., 2023; Kearney and Levine, 2026). An important consideration in these decisions is uncertainty about the future demands of parenthood, given that having a child is an irreversible decision (de la Croix and Pommeret, 2021). In particular, childhood disability can entail substantial caregiving demands and adversely affect parents' employment and mental health (Cheung et al., 2025; Asuman et al., 2025; Chen et al., 2026), and anticipating these consequences may discourage childbearing even before a child is conceived. Fertility decisions may therefore depend on how people perceive both the likelihood of having a child with a disability and the care needs and costs they might face.

This paper examines this possibility using disability service facilities in Japan, where fertility has remained below replacement since the mid-1970s. These facilities may increase opportunities to see people with disabilities and accompanying caregivers in everyday public spaces. These brief encounters can make disability and potential lifelong care needs more salient, potentially increasing the perceived likelihood of having a child with a disability or the anticipated costs of raising one. This hypothesis is supported by survey evidence and online accounts. Survey evidence from Japan and elsewhere suggests that concerns about a child's health or disability are substantial in childbearing decisions, often comparable in importance to work-related concerns. Online accounts in Japan illustrate a possible channel through which everyday encounters with people with disabilities may shape these risk perceptions and thereby influence childbearing decisions. Motivated by them, I examine whether fertility rates are lower near these facilities and whether this negative association is stronger where prospective mothers are more likely to encounter people with disabilities and their caregivers.

I begin by testing whether fertility rates are lower in areas closer to the disability service facilities, which would provide more opportunities for the incidental social contact. I use Japanese data, whose spatial detail and variation in transport use make it possible to examine this relationship at the scale of everyday encounters. Specifically, I use Japan's 2020 Population Census to define the child–woman ratio (CWR), the number of children ages 0–4 per 1,000 women ages 15–49. Measured in 250m grid cells, the CWR, a common proxy of fertility, permits comparisons over distances of a few city blocks. Combining this local measure of the fertility with geocoded facilities from the 2021 National Land Numerical Information Welfare Facilities Data, I estimate distance-band profiles within 1.5km of the nearest facility. The preferred specification includes additive nearest-facility and parent-500m fixed effects and 15 baseline controls to compare nearby areas with similar observed characteristics. The CWR is 7.76 points lower within 250m of a facility than 1.25–1.50km away ($p < 0.01$). Interpreted causally, the nearest-band estimate implies a fertility shortfall equivalent to approximately 1 percent of Japan's births. Including possible spillovers in the 250–500m and 500–750m bands raises this figure to approximately 1.8 percent.

I examine two further predictions about opportunities for incidental social contact. First,

residents along walking routes between facilities and railway stations may be more likely to encounter facility users and caregivers. The actual-path coefficient is significantly negative after controlling for facility distance and three placebo paths constructed by rotating station destinations. Second, private-car travel may reduce encounters on neighborhood streets. The estimated CWR difference between cells within 250m of a facility and those 1.25–1.50km away is −13.05 at the 10th percentile of local car use (6.9 percent), compared with −1.75 at the 90th percentile (79.2 percent). The 11.30-point contrast suggests a stronger negative proximity association where car use is lower. The negative station-path association also weakens as private-car use rises.

I also assess alternative explanations for the lower CWR near disability service facilities. I begin by conducting two placebo tests to assess the possibility that the lower fertility around the disability service facility captures some unobserved local characteristics, rather than the effect of the facility. In the first placebo test, I find that the 2015 CWR was not significantly lower around candidate future facilities observed in 2021 after screening out nearby or text-matched welfare facilities in both the 2011 and 2015 records. This provides no evidence of a comparable negative pre-existing distance gradient around these candidate locations. In the second placebo test, the fully controlled distance and path models do not show significantly lower CWR around elderly care facilities, which primarily address care needs associated with aging. I also examine three alternative explanations behind the lower CWR: migration, land price, and safety. First, although the CWR is biased downward if families with more children tend to move out on net, controlling for the local net migration rates of children only modestly attenuates the lower CWR estimate around the disability service facilities. Land prices and safety levels could also respond to the presence of the disability service facilities while directly affecting the fertility decisions, but conditioning on measured land prices or theft rates retains the lower CWR. Taken together, these checks leave most of the negative association intact.

Overall, these patterns suggest that anticipated caregiving demands and uncertainty about available support may enter fertility decisions. While my analysis focuses on the role of incidental social contacts as a determinant of this anticipation, it also motivates future research on disability-related expectations, which could be shaped by education, mass media, friendship networks, social media, and other factors. These findings also suggest a potential *ex ante* insurance value of public disability support. If concerns about future caregiving demands discourage childbearing, confidence that financial assistance and care services will be available when needed could reduce the burdens and uncertainty prospective parents expect to face. This reassurance could benefit even people who never ultimately use these services, leading to a higher fertility rate. For example, Einav et al. (2025) highlight that the economic cost of having a child with a disability in Sweden could be small under its generous welfare system. Such a generous welfare system for families with a child with a disability certainly helps them, but it may also benefit the entire families beyond them as insurance and lead to a higher fertility rate.

This paper relates to several strands of literature. First, the paper relates to evidence that

reproductive behavior responds to risks to a child's health and to information about those risks. For example, research on the Zika epidemic documents reductions in pregnancies and births following public information linking infection during pregnancy to microcephaly and other congenital conditions (Lautharte and Rasul, 2026; Rangel et al., 2020). Evidence from Japan's rubella epidemic and from reproductive responses after Chernobyl likewise links fertility timing to concerns about fetal health (Mizumoto and Chowell, 2020; Bertollini et al., 1990). These studies highlight both health hazards and the information through which households perceive them. Avoiding fertility during the epidemics relating to the congenital disability is consistent with the complementary literature documenting the economic and welfare costs of raising children with disabilities (Chen et al., 2026; Cheung et al., 2025; Asuman et al., 2025; Frijters et al., 2009; Yamaoka et al., 2016; Skoy, 2024).[1] I extend this perspective to everyday neighborhood exposure, above and beyond a public-health crisis. Indeed, unlike the public-health crisis, a nearby disability facility does not raise the biological probability of having a child with a disability, but it may make that possibility and anticipated caregiving needs more salient. In this sense, this paper sheds new light on the role of the *perceived* risk of having a child with a disability in fertility decisions, even in a usual environment without a public-health crisis.

Second, more broadly, this paper relates to the determinants of fertility. The standard economics approach views childbearing as a household's utility maximization behavior, influenced by income, the costs of raising children, and the allocation of resources between the number of children and investment in each child (Becker, 1960; Becker and Lewis, 1973). In addition to the role of income (Black et al., 2013), recent work emphasizes the compatibility of careers and family life, including childcare provision, family policy, fathers' participation in care, social norms, and flexible labor markets (Doepke et al., 2023; Fernández et al., 2026).[2] Information also matters: experimentally provided information about maternal health risks changes subsequent pregnancy outcomes (Ashraf et al., 2026), while correcting beliefs about pregnancy probabilities changes stated contraceptive intentions (Miller et al., 2025). I bring disability-related expectations about the probability or the caregiving cost of having a child with disability. I test their relevance by a new research design that exploits heterogeneous spatial exposure to people with disability and their caregivers: Examining whether fertility rates are lower near disability service facilities, which would make more salient the anticipated probability and caregiving demand for having a child with disability by providing more opportunities to incidentally encounter people with disability and their caregivers.

Finally, the paper relates to research on how social contact and information shape people's belief and behavior toward another social group. Contact theory raises the possibility that superficial encounters, such as seeing someone on a neighborhood street, leave existing stereotypes

[1]The large cost in Japan is also suggested by disability-related caregiving being an explicit policy concern, with a 2014 government advisory report calling for respite care and support for the employment and psychological well-being of parents raising children with disabilities (Ministry of Health, Labour and Welfare, 2014, pp. 26–28). Generally, the cost may depend on the generosity of the welfare system (Einav et al., 2025).

[2]The economics literature of fertility rates in Japan includes Hashimoto (1974), Yamaguchi (2019), Fukai and Toriyabe (2025), Kitao and Nakakuni (2026), and Kondo (2026).

unchallenged or reinforce them (Allport, 1954). Although intergroup contact generally reduces prejudice, its effects depend on the form and quality of interaction (Pettigrew and Tropp, 2006; Lowe, 2021). In the present setting, incidental encounters may make anticipated care needs more salient, while sustained contact could increase familiarity and reduce concern. Local-interaction models about reproductive behavior provide a related account of how encounters shape beliefs and affect the fertility rate. In Munshi and Myaux (2006), individuals update beliefs about their community's reproductive norms from the actions of the people they meet, so different histories of encounters generate different beliefs and contraceptive choices. Kohler (2000) similarly shows how social networks can shape the transition toward lower fertility. Empirically, evidence that a fertility decision spreads among friends (Balbo and Barban, 2014) and that media portrayals of teenage parenthood might affect teen births (Kearney and Levine, 2015) further connects social exposure to reproductive behavior.[3] Unlike these studies, I study neighborhood differences in opportunities for incidental encounters with people with disability and their caregivers as a potential determinant of fertility decisions.

The rest of the paper is organized as follows. Section 2 presents survey evidence on childbearing concerns, discusses how incidental encounters may shape these concerns, and develops my empirical hypotheses. Section 3 describes the data, sample construction, and empirical design. Section 4 presents my main results about the disability service facility and the fertility, provides additional evidence that the incidental social contact seems important, and assesses alternative explanations through placebo analyses and explicitly accounting for migration, land prices, and safety concerns. Section 5 presents additional heterogeneity analyses and robustness checks. Section 6 concludes. The Online Appendix provides further details on the data construction, empirical specifications, and supplementary results.

2 Motivating Evidence and Hypotheses

Concerns about having a child with a disability and the care needs that may follow could discourage childbearing. Incidental encounters with people with disabilities and their caregivers may make these possibilities more salient, and nearby disability service facilities may provide more opportunities for such encounters. I develop this argument in two steps: first, I examine survey evidence on the importance of child-health and disability-related concerns for childbearing intentions; second, I discuss how incidental encounters may shape these concerns and derive two testable hypotheses.

2.1 Disability-related concerns and childbearing intentions

Surveys from Japan and elsewhere suggest that concerns about a child's health or disability are an important consideration in whether people want children. Table 1 summarizes three indepen-

[3] See Jaeger et al. (2020) for a critical reassessment of Kearney and Levine (2015).

dent Japanese surveys and an international survey on these concerns (see Appendix A.12.1 for more details). All of these surveys ask respondents why they do not want an additional child. In all of the Japanese surveys, survey respondents who do not want an additional child are asked to choose all reasons behind it. The child-health concern is listed along with several other reasons such as financial, housing, and work-related concerns. The international survey explicitly refers to illness or disability. The international survey, in contrast, asks survey takers to rate the importance of disability risks on a four-point scale in their fertility decisions.

I start with the three Japanese surveys. In the Nippon Foundation's 2024 survey, 21.5 percent of childless respondents who did not want children or were content to remain childless cited concern about whether a child would grow up healthy. Among married women aged 49 or younger who did not want a child or another child, concern about a healthy birth was reported by 14.3 percent in the 2021 Japanese Panel Survey of Consumers (JPSC) and 13.4 percent in the 2002 survey by the Japan Institute of Life Insurance (JILI).[4] Comparable concerns appear in the 2024 UNFPA/YouGov survey of adults in 14 countries. Fear of having an ill or disabled child was rated as a very or fairly important reason for not wanting children, currently or in the past, by 36 percent of respondents in the United States, 40 percent in Germany, 33 percent in Sweden, 58 percent in South Korea, and 66 percent in Thailand.

Notably, these concerns about children's health are often reported at rates comparable to work-related concerns, such as career costs or the compatibility of work and family life, whose importance the economics literature has emphasized (Doepke et al., 2023; Adda et al., 2017; Fukai and Toriyabe, 2025). Within the Japanese surveys, the work-related reasons in Table 1 were cited by 19.4 percent of respondents in the Nippon Foundation survey, 10.6 percent in the JPSC, and 15.6 percent in the JILI survey, close to the corresponding child-health shares. In the UNFPA/YouGov survey, difficulty reconciling childrearing with paid work was rated very or fairly important by 37 percent in the United States, 39 percent in Germany, and 68 percent in Thailand, compared with 36, 40, and 66 percent for fear of having an ill or disabled child. These within-survey comparisons suggest that concerns about a child's health or disability warrant attention alongside work-related constraints in understanding childbearing intentions.

I analyze microdata from the JPSC and JILI surveys and find a common pattern: child-health concerns are more frequently reported by women with no children or one child than by those with two or more children. Among married women aged 49 or younger who did not want a child or another child, the corresponding shares are 21.7 versus 12.0 percent in the 2021 JPSC and 24.6 versus 10.5 percent in the 2002 JILI survey (the lower-child-count groups are reported in parentheses in Table 1).[5] This pattern differs sharply from financial or work-related reasons,

[4] In the Nippon Foundation survey, the share reporting child-health concerns is higher among women than men within each age group, and higher among women aged 26–45 than those aged 15–25, although it does not increase monotonically with age (see Appendix A.12.1). One possible interpretation is that these broad child-health responses partly reflect concerns about congenital conditions whose biological risks rise with maternal age, such as Down syndrome (American College of Obstetricians and Gynecologists, 2022).

[5] Adjusted estimates also show a negative association in both JILI and pooled JPSC observations (Appendix Table A.2, Panel A).

Table 1: Survey Evidence on Reasons for Not Wanting Children

Survey	Country	Respondents in the reported denominator	Response format	Child-health concern	Work-related reason (reference)
Nippon Foundation, 2024	Japan	Childless men and women aged 15–45 who did not want children or were content to remain childless; nationwide web survey (n = 1,580).	Multiple response: select all applicable reasons.	21.5%	19.4% Difficulty combining work and childcare.
JPSC, 2021	Japan	Married women aged ≤ 49 who did not want a child or another child; nationwide panel (n = 596).	Multiple response: select all applicable reasons.	14.3% (21.7% for women with ≤ 1 child)	10.6% (8.7% for women with ≤ 1 child) Wanting to continue working.
JILI, 2002	Japan	Married women aged 20–49 who did not want a child or another child; Greater Tokyo registered-panel mail survey (n = 590).	Multiple response: select all applicable reasons.	13.4% (24.6% for women with ≤ 1 child)	15.6% (15.6% for women with ≤ 1 child) Risk of being unable to work.
UNFPA/ YouGov, 2024	United States Brazil Germany Hungary India Indonesia Italy Mexico Morocco Nigeria South Africa South Korea Sweden Thailand	Adults aged 18 or older, including parents; full country samples in an online panel survey (n = 528–1,123 per country).	Rate each reason on a 4-point importance scale: very important; fairly important; not very important; not important at all. Also: don't know; prefer not to say.	15% / 36% 22% / 48% 15% / 40% 23% / 56% 22% / 48% 27% / 48% 17% / 51% 25% / 50% 26% / 44% 37% / 74% 26% / 51% 20% / 58% 12% / 33% 28% / 66%	14% / 37% 25% / 59% 13% / 39% 13% / 44% 21% / 50% 19% / 49% 15% / 54% 24% / 57% 23% / 53% 31% / 73% 20% / 51% 25% / 71% 6% / 24% 26% / 68% Raising a child is too difficult to reconcile with paid work.

Notes: The surveys are the Nippon Foundation's *Survey of Attitudes toward the Declining Birthrate*; the *Japanese Panel Survey of Consumers* (JPSC); the *Survey on Married Women's Life Planning* by the Japan Institute of Life Insurance (JILI); and the 2024 United Nations Population Fund (UNFPA)/YouGov survey of adults in 14 countries. Both percentage columns use the same denominator and response format within each survey and subgroup. Japanese shares report selection of each reason. JPSC and JILI figures are unweighted calculations from microdata. Main figures refer to all eligible women (JPSC: n = 596; JILI: n = 590); parentheses refer to those with one child or less (JPSC: n = 138; JILI: n = 122). Their denominators include reason nonresponse. YouGov's weighted pairs report very important / very or fairly important; denominators include don't know and prefer not to say, and ratings concern reasons for not wanting children now or previously. Child-health items concern healthy development (Nippon Foundation), healthy birth (JPSC and JILI), or illness or disability (YouGov). These responses do not measure forgone births or causal effects. Sources, wording, sample details, and microdata methods are in Appendix A.12.1.

two prominent reasons for not having a child. In both surveys, financial concerns show the opposite contrast: education and child-rearing expenses were cited by 30.4 versus 44.8 percent in the JPSC and 48.4 versus 67.5 percent in the JILI survey. Work-related reasons differ little between the two groups: 8.7 versus 11.1 percent in the JPSC and 15.6 percent in both groups in the JILI survey. Together, these patterns suggest that concerns about a future child's health may be particularly salient among women with no or few children, above and beyond the financial and work-related constraints extensively studied in the fertility literature (Kearney and Levine, 2026).

Taken together, these surveys suggest that concerns about a future child's health or disability play an important role for stated childbearing intentions. However, they do not establish a effect of these concerns on realized fertility or explain how the concerns are formed. I next consider incidental encounters with people with disabilities and their caregivers as one potential influence on these expectations, and how it may lead to a lower ferility rate.[6]

2.2 Incidental encounters and disability-related expectations

Online accounts illustrate how individuals connect everyday encounters with people with disabilities and their caregivers to concerns about their own future children. I conducted a purposive, non-systematic online search of publicly available Japanese-language discussion boards, question-and-answer sites, and edited first-person or interview-based web articles to find accounts consistent with this proposed link (see Appendix A.12.2 for more details). The English translations below retain the speakers' wording.

(1) "The mothers I see accompanying people around town […] when I think there is a chance I might have a child with an intellectual disability, I cannot have children." *(Comment on GirlsChannel, a Japanese discussion forum for women, 2018)*

(2) "I want children, but I really do not want a child with a disability. Even on the train, when I see parents accompanying people with disabilities, I think how hard it must be for them—or rather, it makes me sad… I do not have the heart of a saint… I cannot give birth unless I am 100% certain that my child will not be born with a disability." *(Comment on GirlsChannel, a Japanese discussion forum for women, 2017)*

(3) "I sometimes see people accompanying children with profound disabilities […] I am not confident that I could raise my own child if they were in that condition." *(Answer on Yahoo! Chiebukuro, a Japanese question-and-answer site, 2026)*

[6] Although the reasons are elicited only from women who report not wanting a child or another child, the JPSC panel allows me to test whether selecting the child-health reason predicts subsequent births even among women who do not want a child. Among married women aged 24–39 at baseline, selecting this reason is associated with a statistically significant 1.23 percentage-point lower probability of a birth in the following year relative to not selecting it, after adjusting for age, existing family size, and other baseline characteristics (Appendix Table A.2, Panel B). Therefore, although it misses an important margin that the health concern may make women not wanting to have a child, the JPSC data are consistent with this paper's results in suggesting the negative fertility impact. More details are provided in Appendix A.12.1.

These accounts link seeing people with disabilities and their caregivers in public spaces to anticipated care needs, concern about having a child with a disability, and hesitation about childbearing. They illustrate a possible connection between everyday encounters and childbearing concerns, rather than establish the effects of encounters or proximity to a facility.

Standard behavioral theories of availability and salience provide a theoretical basis for this connection. Seeing people with disabilities does not change the biological probability that one's child will have an innate disability, but it may make that possibility and its perceived consequences easier to imagine.[7] Readily recalled examples can influence subjective probability judgments (Tversky and Kahneman, 1973), while salient outcomes can receive greater weight in decisions under risk (Bordalo et al., 2012). Repeated incidental encounters may therefore make the possibility of raising a child with a disability more readily come to mind and draw attention to potential long-term care needs, financial pressure, and implications for other family members. Such encounters could influence childbearing decisions by increasing the perceived likelihood of these outcomes or, even without changing perceived likelihoods, by increasing the weight placed on their anticipated consequences.[8]

Disability service facilities provide a setting in which to examine this proposed mechanism. Recent evidence using GPS data shows how the places people visit shape opportunities for social interaction (Atkin et al., 2022; Nilforoshan et al., 2023; Couture et al., 2025; Massenkoff and Wilmers, 2025). Facilities bring users and their caregivers to particular locations, potentially increasing opportunities for incidental encounters in surrounding public spaces. If these encounters increase perceived disability risks or the anticipated burden of care, fertility should be lower where such encounters are more likely.[9] This reasoning yields the first hypothesis:

Hypothesis 1 (facility proximity and fertility rates). Local fertility is lower near disability service facilities, with the largest difference in the immediate vicinity and attenuation over distance.

Proximity is just one proxy for contact opportunities. Residents along walking routes between facilities and railway stations may also encounter users and caregivers more often, while private-car travel may reduce such encounters. These differences imply a second hypothesis:

[7] More sustained or favorable contact could instead reduce unfamiliarity and concern (Pettigrew and Tropp, 2006; Lowe, 2021). However, the practicum account in Appendix A.12.2 illustrates that sustained contact may also make anticipated care needs more salient and enter childbearing considerations.

[8] The irreversibility of childbearing makes anticipated care needs relevant before a child is conceived (de la Croix and Pommeret, 2021). It also motivates a possible insurance value of support for people with disabilities and their caregivers, discussed again in Section 6.

[9] Nearby facilities may also inform prospective parents about available support for people with disabilities. This reassurance could reduce anticipated caregiving burdens or uncertainty and encourage childbearing, potentially offsetting the proposed discouraging effect of incidental encounters.

Hypothesis 2 (heterogeneity by opportunities for incidental social contact). Conditional on facility distance, fertility is lower along walking routes between facilities and railway stations. The negative proximity association is stronger where private-car use is lower.

3 Data and Empirical Design

To test the hypotheses 1 and 2 presented in Section 2, this section describes the data and empirical strategy used in this paper. I start with describing the census data, the fertility measure, and the construction of spatial exposure to disability service facilities. I then present the empirical specification. Appendix Table A.1 reports summary statistics for the main estimation sample. Full source citations, file identifiers, and construction details are provided in Appendix A.

3.1 Census mesh data and the fertility outcome

I use Japan's 2020 Population Census regional-mesh statistics. Regional meshes form a standardized latitude–longitude grid whose boundaries are independent of administrative divisions. Figure 1(a) shows the nationwide grid schematically and enlarges a parent 500m mesh to illustrate its four 250m cells arranged in a two-by-two grid.[10] Each 250m cell is a unit of observation in the main analysis, while the parent 500m mesh defines the local neighborhood, which I account for by the fixed effects in Section 3.3.

Using population counts by sex and five-year age group in each 250m cell, I define the main dependent variable for cell i, the child-to-woman ratio (CWR), as

$$\mathrm{CWR}_i^{15\text{–}49} = 1000 \times \frac{\text{children ages 0–4}_i}{\text{women ages 15–49}_i}. \tag{1}$$

The numerator includes children aged between 0 and 4 of both sexes, while the denominator is the number of women of childbearing age.[11] The CWR is a commonly used measure of fertility rates at a spatially granular level, but note that the CWR is different from an annual birth rate or a measure of completed fertility. In particular, the CWR could be influenced by migration behavior that is not directly related to reproductive behavior. Section 4.3.2 examines the sensitivity of my results to migration.

I combine census counts across all 47 prefectures. To preserve the correspondence between population counts and individual mesh areas, I retain cells with a strictly positive number of women ages 15–49.[12]

[10] I call the 500m mesh that contains the 250m mesh of interest the "parent" mesh. I also use the words "mesh" and "cell" interchangeably.

[11] Following the standard definition of the CWR, I define the childbearing age as 15–49. Using the 20–39 definition does not change my conclusion (Section 5).

[12] Some cells with few observations are omitted because the census sometimes refrains from reporting the relevant variable to secure confidentiality.

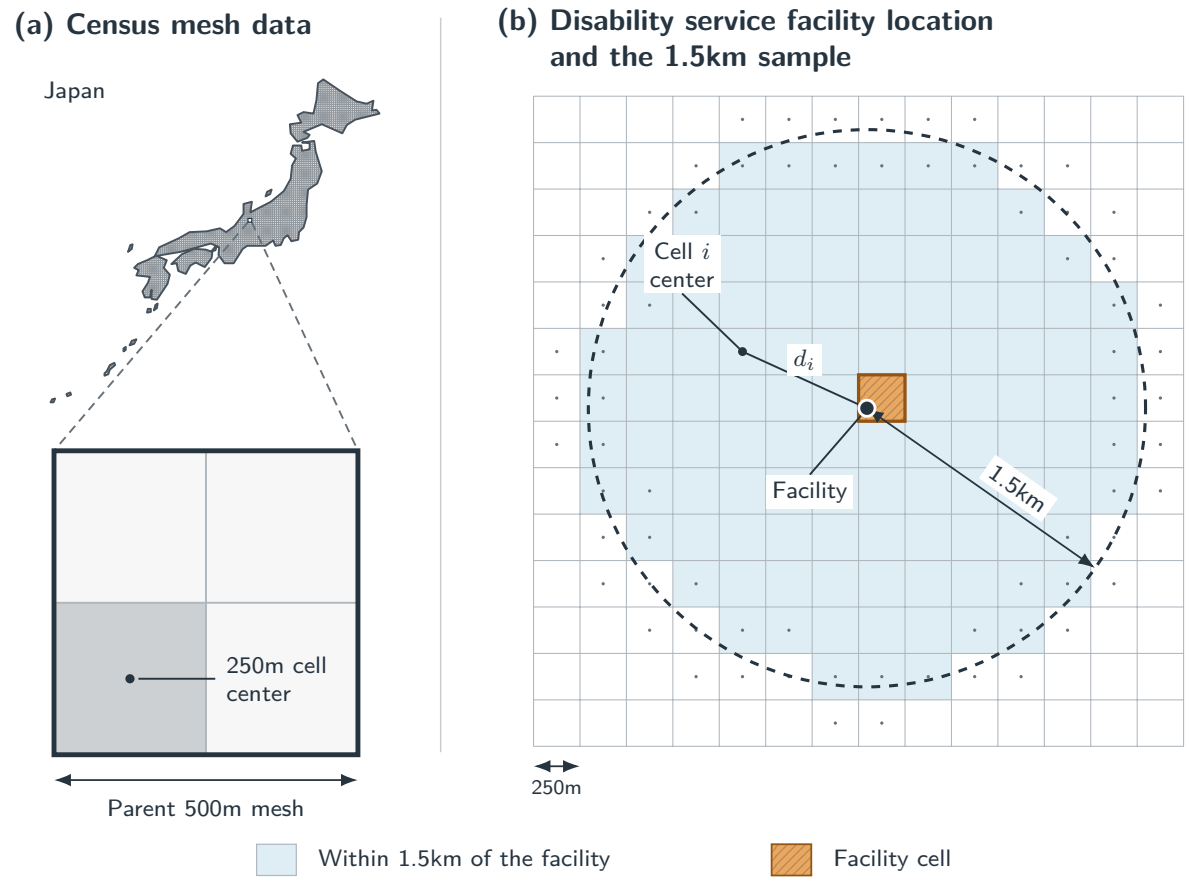


Figure 1: Census Meshes, Facility Locations, and the Spatial Sample

Notes: Panel (a) shows how each 500m mesh divides into four 250m cells. In panel (b), blue cells lie within 1.5km of the facility, measured from their centroids; the orange hatched facility cell is included. The dashed circle marks the 1.5km cutoff, and d_i is cell i's distance to the facility. The analysis assigns each cell to its nearest disability service facility and applies the additional sample restrictions in the text. The maps are schematic and not to scale.

3.2 Facility locations and spatial exposure

Facility location data come from the 2021 National Land Numerical Information Welfare Facilities Data. These data comprehensively cover a range of residential and nonresidential disability services, such as residential support facilities and group homes, day-service and employment-support establishments, and consultation offices (Ministry of Health, Labour and Welfare, 2018). I pool all facility types in the analysis because most records fall into a broad category without service-specific distinctions (see Appendix A.3 for details).

Figure 1(b) illustrates the following sample selection procedure. I assign each 250m cell in the Census to its nearest admissible disability service facility using great-circle distance from the cell centroid.[13] I keep cells less than 1.5km away and group distances into six 250m bands, with

[13]Distances to the nearest facility tend to be shorter in urban areas with more closely spaced facilities. However, after conditioning on both sets of fixed effects, the 15 baseline controls, and the other distance-band indicators, the residualized variation identifying the short- and long-distance coefficients has a broadly similar composition across population-density groups. See Appendix Figures B.1 and B.2. Section 5 also examines the number of facilities within 250m and finds a stronger negative CWR association as this number increases.

1.25–1.50km as the reference.[14] Facility-containing cells are included in the main sample.[15]

3.3 Empirical specification and identification

Let $f(i)$ denote the nearest admissible facility assigned to cell i, $m(i)$ its unique parent-500m mesh, and D_{ib} the five distance-band indicators: 0-250m, 250-500m, 500-750m, 750-1000m, 1000-1250m. The distance of 1250-1500m is used as the reference category.

The specification for the main analysis in Section 4.1 is:

$$\mathrm{CWR}_i = \sum_{b=0}^{4} \beta_b D_{ib} + X_i'\gamma + \alpha_{f(i)} + \delta_{m(i)} + u_i. \tag{2}$$

The coefficient β_0 in Equation (2) compares the 0–250m and 1.25–1.50km bands, which is the main interest. Analogously, $\beta_1, \beta_2, \beta_3, \beta_4$ capture the fertility differential for cells that are away from the nearest disability facility by, respectively, 250–500m, 500–750m, 750–1000m, and 1000–1250m. Here, nearest-facility fixed effects $\alpha_{f(i)}$ absorb differences across facility catchments, while parent-500m fixed effects $\delta_{m(i)}$ absorb characteristics shared by adjacent census cells.

X_i contains the following 15 baseline controls: log population density; six female age shares (20–24, 25–29, 30–34, 35–39, 40–44, and 45–49, each divided by women ages 15–49, with 15–19 omitted); the university-graduate share; three zoning indicators; three housing-form shares; and log distance to the nearest station. The education denominator pools school graduates ages 15 or older of both sexes. Housing controls measure row houses, apartments, and other or unstated forms, with detached houses omitted. Cells outside designated zoning polygons have all three zoning indicators equal to zero. The distance to the nearest station is the shortest road distance (see Appendix A.6 for calculation details). Appendix A.5 gives detailed variable definitions.

In my estimation sample, I require complete observations for the outcome and all 15 baseline controls. To avoid perfect collinearity with fixed effects, I then recursively remove cells that are the sole remaining observation in either their nearest-facility group or their parent-500m mesh. The resulting main estimation sample contains 331,850 cells assigned to the nearest facility among 36,377 facilities.

The baseline specifications are estimated by weighted least squares using the number of women ages 15–49 as analytic weights, aiming to replicate all women at the childbearing age in Japan as the population. Standard errors in Equation (2) and all columns of Table 2 are clustered two ways by nearest facility and parent-500m mesh (Cameron et al., 2011).

[14] Using cells within 2km from the nearest facility does not change my results (see Section 5).

[15] Since residential services could mechanically affect the local population composition when there are residents of the facilities, I also estimate the model excluding these cells. The results do not change (see Section 5).

4 Main Results

4.1 Proximity to a disability service facility and the fertility rate

Table 2 brings together the distance, station-path, and private-car-use results. The coefficients of distance bins (i.e., β_b) in Columns (1)–(2) are also plotted in Figure 2. Both include facility and parent-500m fixed effects; column (2) adds the 15 baseline controls. Columns (3)–(5) report the extensions discussed in the following subsections.

Table 2 and Figure 2 show the nearest-band coefficient is negative and statistically distinguishable from zero in both distance-only specifications. Without controls, column (1) gives an estimate of -6.711 ($p = 0.028$). In the preferred specification in column (2) with control variables, cells within 250m have 7.76 fewer children ages 0–4 per 1,000 women ages 15–49 than cells 1.25–1.50km away ($p = 0.001$). In column (2), the 250–500m and 500–750m estimates are -4.14 ($p = 0.072$) and -3.65 ($p = 0.099$). Both are significant at the 10 percent, but not the 5 percent, level; the farther bands are statistically indistinguishable from zero even at the 10 percent level. The negative estimates in these two bands suggest possible spillovers beyond 250m, although the strongest evidence concerns the nearest band.

Interpreted causally, the 0–250m estimate implies a fertility shortfall equivalent to approximately 1 percent of Japan's births.[16] Including possible spillovers in the 250–500m and 500–750m bands raises this figure to approximately 1.8 percent. Given that Japan's total fertility rate (TFR) was approximately 1.4 in the study period, this implies an impact on Japan's TFR of around 0.015–0.025. These calculations aggregate the estimated gaps over the corresponding estimation-sample cells, relative to the 1.25–1.50km reference band.

4.2 Opportunities for incidental social contact

Above and beyond distance to a disability service facility, Hypothesis 2 highlights how the fertility rates may be related to opportunities to see people with disabilities and accompanying caregivers on neighborhood streets. I examine two complementary features of this environment: modeled walking routes to railway stations and neighborhood reliance on private cars.

4.2.1 Paths between Disability Service Facilities and Train Stations

To evaluate the route prediction in Hypothesis 2, I use modeled walking routes between disability-service facilities and railway stations. For each facility, I choose the minimum road-access distance, including the endpoint connections, among the five nearest station groups. This destination need not be the closest station in straight-line distance. I then define the actual-path indicator, which equals one when a cell's rectangle intersects an available path assigned to its nearest facility, as indicated in Figure 3(a). The indicator is missing if the station-derived path is unavailable.

[16] Appendix A.2 explains more details behind this calculation.

Table 2: Regression results: Disability service facility and the CWR

	Distance		Station	Car-use heterogeneity	
	No controls	Controls	paths	Distance	Distance + paths
	(1)	(2)	(3)	(4)	(5)
A. Distance-band and station-path coefficients					
0–0.25km	−6.711**	−7.757***	−6.129**	−7.710**	−6.872**
	(3.062)	(2.380)	(2.466)	(3.121)	(3.219)
0.25–0.50km	−2.935	−4.140*	−3.758	−4.471	−4.768
	(2.976)	(2.305)	(2.344)	(3.058)	(3.122)
0.50–0.75km	−2.830	−3.646*	−3.372	−3.661	−4.062
	(2.858)	(2.212)	(2.248)	(2.991)	(3.051)
0.75–1.00km	−1.848	−2.801	−2.543	−2.637	−2.991
	(2.571)	(1.996)	(2.028)	(2.761)	(2.818)
1.00–1.25km	−0.264	−0.590	−0.442	0.564	0.453
	(1.953)	(1.556)	(1.576)	(2.091)	(2.120)
Actual station path	—	—	−2.118***	—	−2.030***
			(0.686)		(0.681)
Placebo path: 90°	—	—	−0.910	—	−0.879
			(0.634)		(0.626)
Placebo path: 180°	—	—	1.326**	—	1.268**
			(0.635)		(0.623)
Placebo path: 270°	—	—	−1.126*	—	−1.009
			(0.631)		(0.622)
B. Conditional CWR coefficients at private-car-use percentiles					
0–0.25km distance band					
P10 (low car use)	—	—	—	−13.055**	−12.582**
				(6.048)	(6.174)
P50 (median car use)	—	—	—	−8.262**	−7.546**
				(3.368)	(3.504)
P90 (high car use)	—	—	—	−1.755	−0.380
				(3.084)	(3.260)
Actual station path					
P10 (low car use)	—	—	—	—	−5.143***
					(1.162)
P50 (median car use)	—	—	—	—	−2.397***
					(0.703)
P90 (high car use)	—	—	—	—	1.509
					(1.085)
Distance interactions: joint p	—	—	—	< 0.001	0.018
Path interactions: joint p	—	—	—	—	< 0.001
Facility and parent-500m FE	Yes	Yes	Yes	Yes	Yes
Controls	No	Yes	Yes	Yes	Yes
Observations	331,850	331,850	315,357	331,850	315,357

Notes: The outcome is the 2020 CWR. The reference distance band is 1.25–1.50km. All models use women ages 15–49 as weights and two-way clustering by facility and parent-500m mesh (standard errors in parentheses). Column (1) has no controls; columns (2)–(5) include all 15 controls. Columns (1), (2), and (4) share the distance-analysis sample; columns (3) and (5) share the smaller sample with all four station and four bus paths available, but include only station paths. In columns (4)–(5), Panel A coefficients are evaluated at each sample's women-weighted mean car use (41.086 percent in column (4) and 40.506 percent in column (5)); Panel B reports the 0–250m coefficient in both columns (4) and (5), and the actual-path coefficient in column (5), evaluated at P10, P50, and P90 of each sample's women-weighted car-use distribution. These rates are 6.9/37.6/79.2 percent in column (4) and 6.8/36.5/78.8 percent in column (5); standard errors use the full covariance matrix. The car-use level is absorbed by parent-500m FE. Dashes indicate terms or summaries not applicable to that column. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

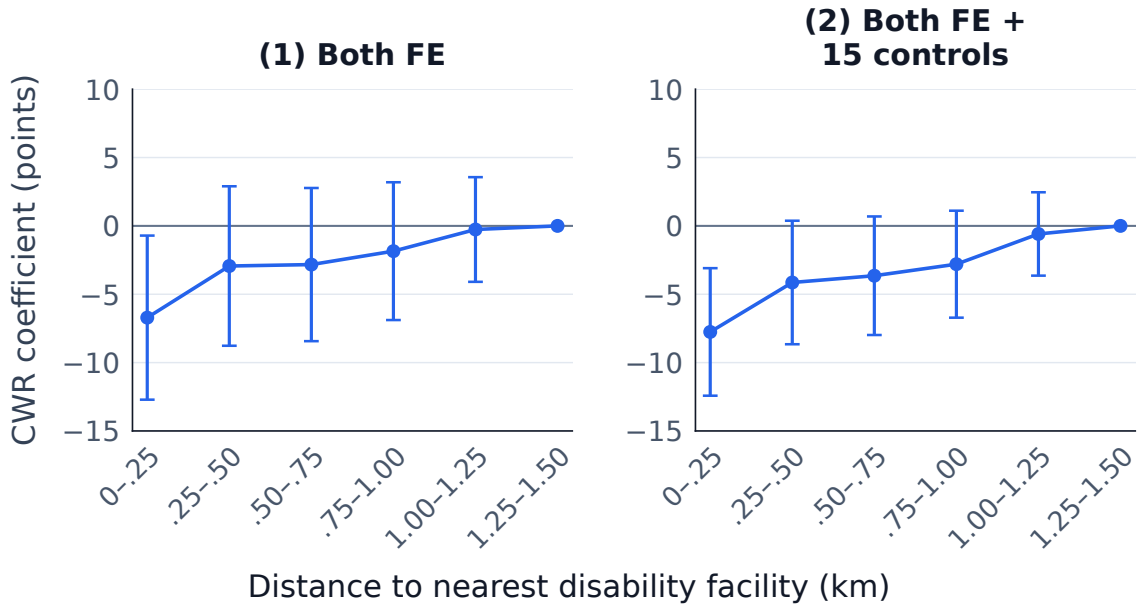


Figure 2: CWR and the Distance from the Nearest Disability Facility: Graphical Illustration

Notes: Points show the distance-band coefficients in columns (1)–(2) of Table 2, relative to 1.25–1.50km; bars show 95 percent confidence intervals. CWR is children ages 0–4 per 1,000 women ages 15–49. Both specifications use the same 331,850 cells, weighted by women ages 15–49, with facility and parent-500m fixed effects and two-way clustered standard errors. Column (1) has no controls; column (2) includes all 15 controls.

To see how the fertility rate along the actual path to the station stands out, I compare the results with the "placebo paths" that lead to the same disability service facility but from hypothetical railway stations. Figure 3(b) illustrates the construction of the placebo paths. I first rotate each railway station around the facility by 90, 180, or 270 degrees, recompute the walking path, and apply the same intersection rule. Rotation preserves the straight-line endpoint distance, while routed lengths can change depending on the road network. See Appendix A.6 for more details of the procedure.

In my analysis, I add the actual-path and three rotated-path indicators to my baseline model (2):

$$\begin{aligned} \text{CWR}_i = \sum_{b=0}^{4} \beta_b D_{ib} + \sum_{q \in \{A,90,180,270\}} \theta_q R_i^q \\ + X_i' \gamma + \alpha_{f(i)} + \delta_{m(i)} + u_i. \end{aligned} \tag{3}$$

Here R_i^A denotes the actual station-path indicator, while R_i^{90}, R_i^{180}, and R_i^{270} denote the rotated-path indicators defined above. Thus, θ_A measures the actual-path CWR difference conditional on radial distance and the other paths. Weighting and clustering are conducted as in Section 3.3. Appendix B.2 provides further details.

Column (3) of Table 2 reports this joint station-path specification, and its actual and placebo path coefficients are also visualized in panel (a) of Figure 4. In this specification, the actual-path

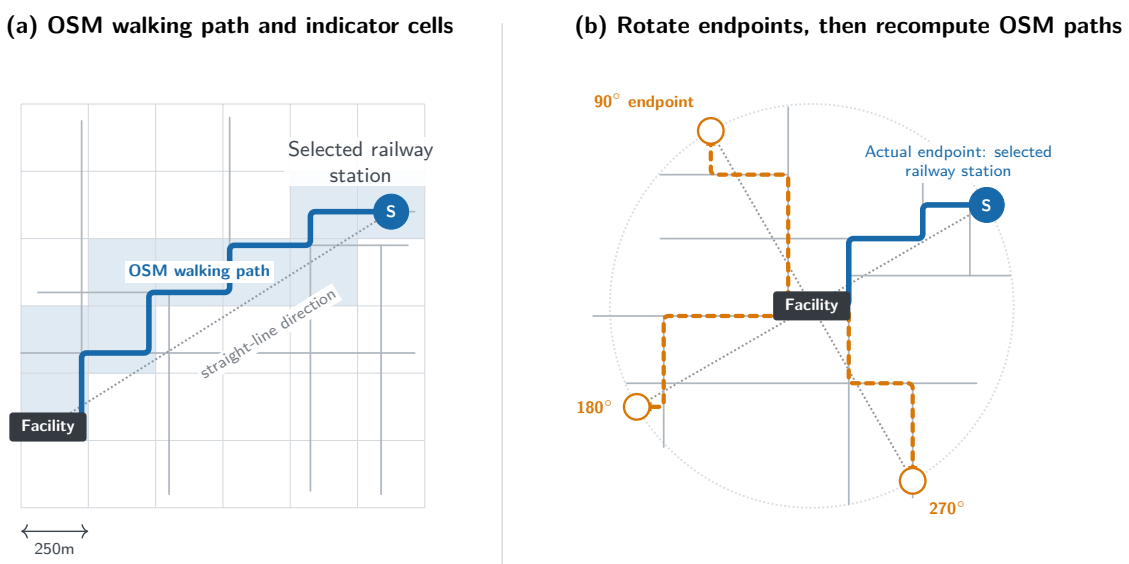


Figure 3: Construction of OSM-Routed Actual and Rotated Transit Paths

Notes: Panel (a) shows a walking path from a facility to its selected railway station. A shaded cell intersects an available path from its assigned facility. Facility-containing cells remain eligible for estimation. In panel (b), the destination is rotated by 90, 180, or 270 degrees around the facility, and the walking path is recalculated. Rotation keeps the straight-line distance unchanged, but the walking distance can change. Appendix A.6 describes path construction and availability. The diagram is not to scale.

coefficient is −2.118 ($p = 0.002$).[17] In contrast, the 90-, 180-, and 270-degree coefficients are −0.910 ($p = 0.151$), 1.326 ($p = 0.037$), and −1.126 ($p = 0.074$). The actual-path coefficient is lower than the mean of the three rotations by 1.881 points. This suggests that consistent with the first part of Hypothesis 2, cells along the path between disability service facilities and train stations may have a lower CWR.

4.2.2 Local private-car use

If private-car travel reduces encounters on streets and around transit stops, the negative CWR association should be weaker where car use is higher. To test this hypothesis, I take the 2020 Population Census commuting-mode counts at the parent-500m mesh level. I use the car use rate in commuting as a proxy of the local car dependence (see Appendix A.5 for the variable construction). I allow the distance profile in the model (2) to vary with the car use rate as follows:

$$\begin{aligned} \mathrm{CWR}_i = &\sum_{b=0}^{4} \beta_b D_{ib} + \sum_{b=0}^{4} \theta_b z_{m(i)} D_{ib} \\ &+ X_i'\gamma + \alpha_{f(i)} + \delta_{m(i)} + u_i. \end{aligned} \tag{4}$$

Here $z_{m(i)} = (c_{m(i)} - \bar{c}_w)/0.10$, where $c_{m(i)}$ is the share of commuting workers and students using private cars, excluding people working at home, and $\bar{c}_w \simeq 0.41086$ is its sample mean with

[17] Replacing station paths with bus-stop paths also yields a negative actual-path coefficient (−2.459, $p = 0.002$) with all controls on the same sample (Appendix Table B.2).

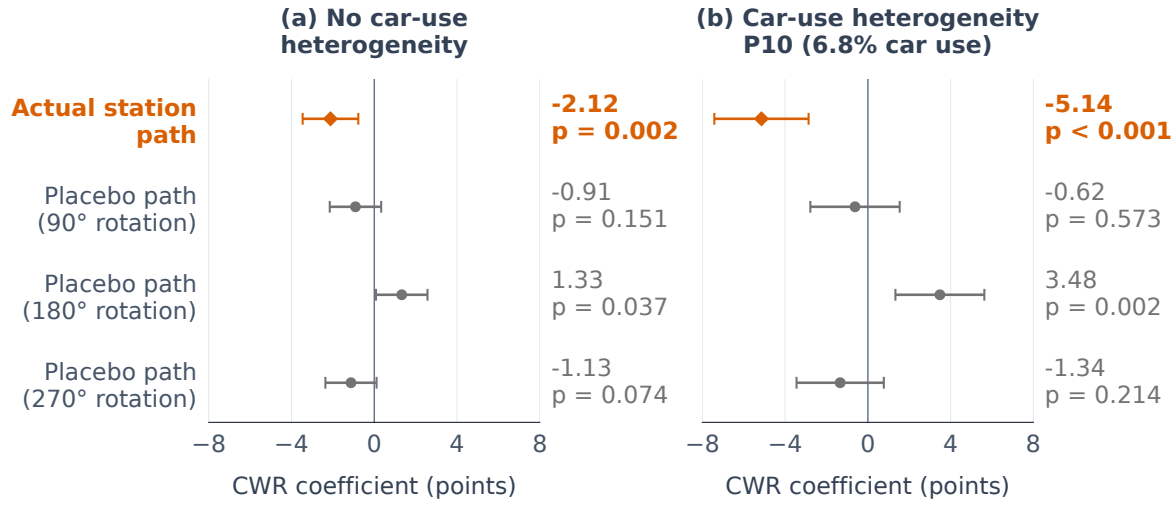


Figure 4: Estimated CWR Differences Along Actual and Placebo OSM Station Routes

Notes: Points show CWR differences for cells intersecting each path, estimated jointly; bars are 95 percent confidence intervals. Both panels include the five distance bands, facility and parent-500m fixed effects, and all 15 controls. Panel (a) has no car-use interactions and corresponds to column (3) of Table 2 and column (2) of Appendix Table B.1. Panel (b) interacts all five distance bands and all four station paths with car use, as in column (5) of Table 2, and evaluates the path coefficients at the sample's women-weighted P10 of car use (6.8 percent).

the number of women at the childbearing age as weights. An increase of one in $z_{m(i)}$ therefore represents 10 percentage points more car use. Because the car-use rate is measured at the parent 500m mesh level, its direct effect is absorbed by the parent-500m fixed effects. θ_b measures the change in the band-b CWR difference per 10-percentage-point increase in car use.

Column (4) of Table 2 reports the distance-band main effects at mean car use in Panel A and the nearest-band CWR coefficient at the 10th, 50th, and 90th percentiles of car use in Panel B. The joint test covers all five distance-band interactions retained in the model. Weighting and clustering follow Section 3.3. Figure 5 visually presents the estimates evaluated at the 10th, 50th, and 90th percentiles (6.9, 37.6, and 79.2 percent) of the car-use rate. The nearest-band estimates, also reported in Panel B of Table 2, are, respectively, −13.05, −8.26, and −1.75. Overall, consistent with the second part of Hypothesis 2, we find evidence that the fertility rate is much lower around a disability service facility when car-use rate is low, while no such relationship is observed in a car-dependent society.[18]

We also examine whether the lower CWR along the route between disability service facilities and train stations is more salient in areas with less car-dependence. Column (5) of Table 2 allows both radial-distance and station-path coefficients to vary with car use, and Figure 4(b) shows the path coefficients at the 10th percentile of car use (see Appendix Figure B.3 for other car-use levels). Compared with our baseline result in Figure 4(a), I find that the CWR is much lower along the actual path, while the coefficients of placebo routes hardly change from the

[18]Section 5 examines additional heterogeneity by the cell-level mean age of women ages 15–49 and the local university-graduate share.

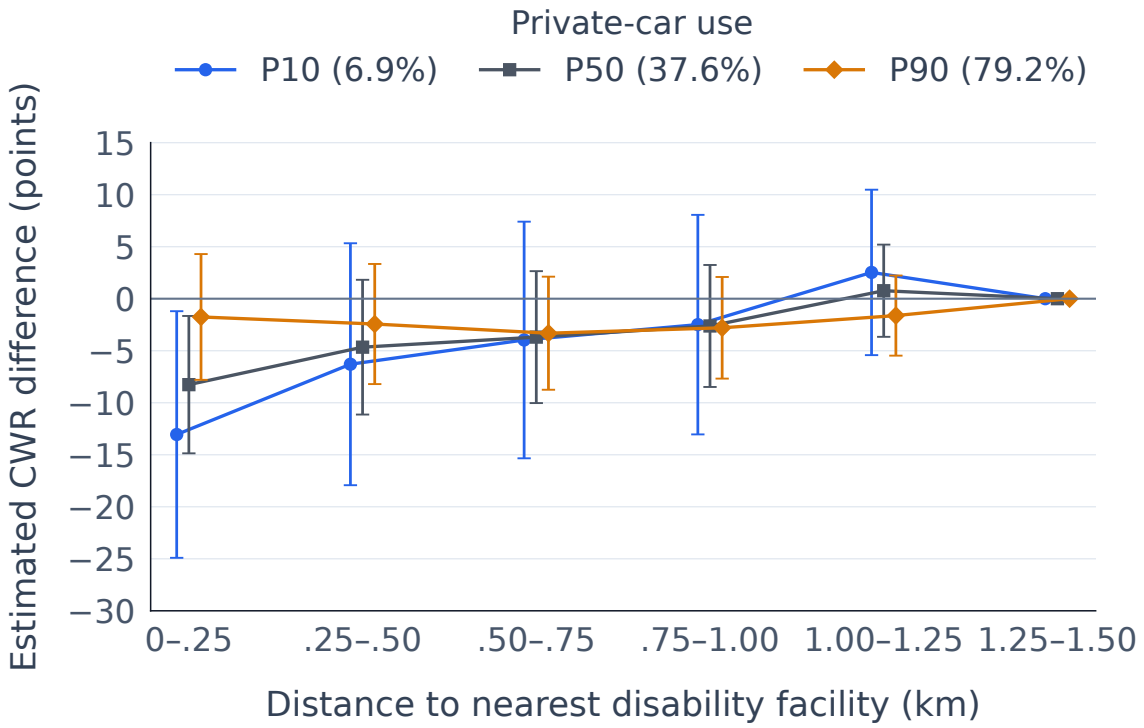


Figure 5: The disability facility and the CWR by private-car-use rate

Notes: The model estimates a separate car-use interaction for each of the five distance bands (Equation 4) and includes facility and parent-500m fixed effects and all 15 controls. It corresponds to column (4) of Table 2. Points compare CWR with the 1.25–1.50km band at the 10th, 50th, and 90th percentiles of car use (6.9, 37.6, and 79.2 percent). Bars are pointwise 95 percent confidence intervals.

baseline. This is again consistent with Hypothesis 2: more opportunities for incidental social contact, such as along facility-station paths when people do not use cars, are associated with a lower CWR.

4.3 Assessing alternative explanations

This section assesses alternative explanations for the lower CWR near the disability facilities in two steps. Section 4.3.1 examines, as placebo tests, the CWR right before the entry of disability facilities and the CWR around elderly care facilities. Section 4.3.2 evaluates how my results might be driven by residential sorting, land prices, and safety level.

4.3.1 Placebo analyses

I use two placebo analyses to assess pre-existing local fertility patterns and proximity to welfare facilities more generally. The first examines whether the 2015 fertility rate was already low in areas that did not yet host a disability service facility but later hosted it in 2021. The second examines whether a similar CWR profile appears around elderly care facilities.

Future entry of disability facilities placebo. A potential concern is that areas that host the disability service facilities have unobserved location characteristics that are correlated with the lower fertility rate. For example, neighborhood opposition can constrain the establishment of disability service facilities in Japan (Okahara et al., 2022). If opposition is stronger in neighborhoods with more young children, facilities may locate in areas with lower pre-existing CWRs.

This placebo asks whether the 2015 CWR was already lower near candidate future facilities. More specifically, this placebo analyzes the relationship between the 2015 CWR and disability service facility records observed in the 2021 facility location data but unobserved in the 2011 and 2015 data.[19] These locations may share characteristics that attract disability services. Under the contact hypothesis, however, a facility would not affect local CWR before it begins operating. I report the distance-only specification with both fixed effects and all available year-specific controls; Appendix B.5.1 provides the specification details.

Figure 6, Panel A, reports the results (see Appendix Table B.4 for the full regression results). The nearest-band estimate is -1.03 ($p = 0.598$), which is substantially smaller in absolute magnitude than -7.76 in Section 4 and statistically indistinguishable from zero.[20] In addition, I also find no evidence that the CWR was already lower along the path between train stations and disability facilities that did not exist as of 2015 but appeared by 2021 (Appendix Table B.5). Overall, I do not find evidence that areas around the disability service facilities already had a lower CWR even before the facilities appeared.[21]

Elderly care facility placebo. If facilities primarily providing care associated with aging had a similar negative effect on fertility, then incidental social contact with people with disability would be less likely to be the primary reason behind the lower CWR. For example, it is possible that areas around welfare facilities tend to have certain location characteristics, such as unobserved zoning regulations or political orientation, that are preferred by those with weaker fertility intentions.

I now check this possibility by examining the impact of the elderly care facilities, which are another form of welfare facilities but likely to bring little direct benefit to young families, on fertility rates. Among the elderly care facilities in my welfare facility dataset, I retain only elderly care facilities strictly more than 1km from any admissible disability service facility to focus on their effects, leaving 15,492 facility points. I then repeat the same analysis as Section 4, replacing disability service facilities with these elderly care facilities.

Panel B of Figure 6 reports the results: the coefficient is 1.246 ($p = 0.647$), and none

[19] Appendix A.7 provides more details about the procedure to construct this future-facility data.

[20] This result is not due to using the 2015 data instead of 2020 because repeating the analysis in Section 4 using the 2015 data yields similar results (Appendix Table B.7).

[21] In contrast, Appendix Table B.6 reports a negative 2020 CWR coefficient around the facilities that entered between 2015 and 2021. Subtracting the 2015 coefficient (Appendix Table B.4) gives a difference-in-differences contrast of -2.711 for the 0–250m band (SE 2.259, $p = 0.230$), which is negative but statistically insignificant. Its smaller magnitude than in Section 4 may reflect entry timing close to or after 2020, limiting opportunities of exposure to the facility users and their caregivers.

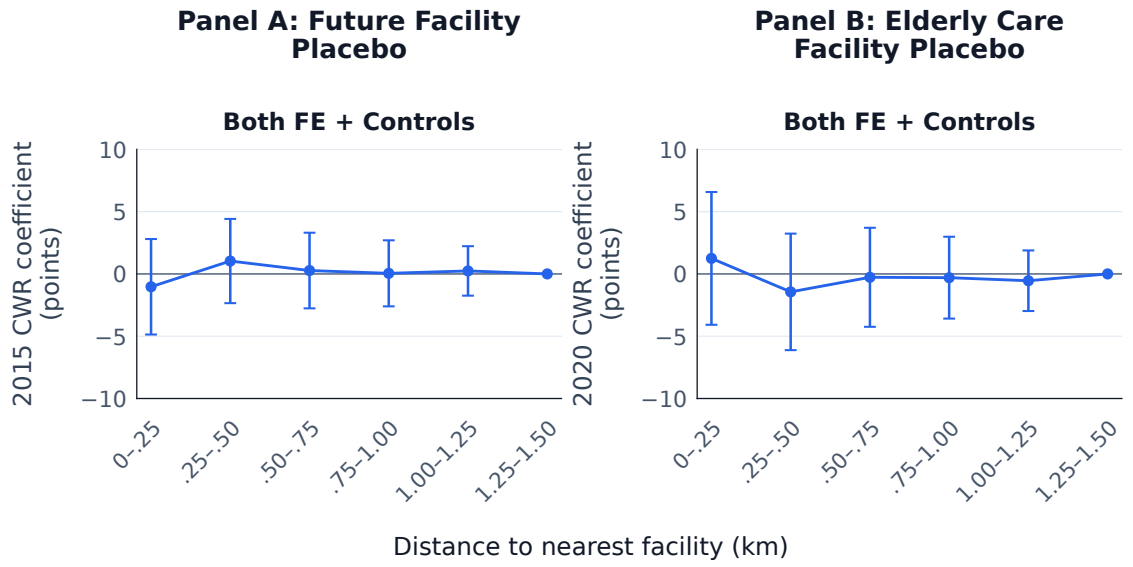


Figure 6: Placebo Tests

Notes: Points show CWR differences from the 1.25–1.50km band; bars are 95 percent confidence intervals. Panel A (left) uses the 2015 CWR around facilities that did not exist as of 2015 but appeared in the 2021 facility data (Appendix A.7; N = 198,147). Panel B (right) uses the 2020 CWR around elderly-care facilities strictly more than 1km from all admissible disability facilities. Panel A corresponds to Appendix Table B.4, and Panel B to column (2) of Appendix Table B.8.

of the five distance coefficients is statistically significant (see Appendix Table B.8 for the full regression results). Moreover, I find no evidence that the CWR is lower along the routes between train stations and the elderly care facilities (Appendix Table B.9). Taken together, the lower CWR is observed only for disability service facilities, not for elderly care facilities.

4.3.2 Accounting for confounders

Although I include facility and parent-500m fixed effects and various controls, unobserved confounders could remain. Here I explicitly consider three potential confounders: migration, land prices, and safety. As discussed below, these factors play, at most, a limited role in explaining the negative association between disability service facilities and fertility.

Migration. An important concern for using the CWR as a fertility measure is that selective migration (i.e., residential sorting) can contaminate it by altering the relative numbers of young children and reproductive-age women without reproductive behavior. Specifically, the CWR falls if households with more young children per woman are more likely to leave or less likely to enter a neighborhood. Therefore, even without any change in childbearing decisions, a change in residential composition could lower the CWR around the disability facility if it selectively attracts those with fewer children.[22]

To address this concern, I construct a proxy of the selective migration as follows. Note first

[22]Note that proportional changes in children and women leave it unchanged, implying that the volume of migration itself does not necessarily translate into the bias in the CWR.

Table 3: Sensitivity to Controlling for Migration, Land Prices, and Safety Level

Distance band (km) / regressor	(1)	(2)
Panel A: Net child migration rate		
0–0.25	−6.771**	−7.861***
	(3.279)	(2.539)
0.25–0.50	−3.705	−4.749*
	(3.197)	(2.474)
0.50–0.75	−3.515	−4.315*
	(3.088)	(2.386)
0.75–1.00	−2.802	−3.581*
	(2.805)	(2.169)
1.00–1.25	−0.633	−1.414
	(2.166)	(1.727)
1.25–1.50 (reference)	0	0
Net child migration rate (5–9 as of 2020)	0.077***	0.042***
	(0.027)	(0.014)
Net child migration rate (10–14 as of 2020)	−0.001	−0.006
	(0.011)	(0.005)
Observations	240,193	240,193
Panel B: Residential land price		
0–0.25	−6.710**	−7.330***
	(3.068)	(2.382)
0.25–0.50	−2.935	−3.781
	(2.980)	(2.307)
0.50–0.75	−2.830	−3.378
	(2.862)	(2.215)
0.75–1.00	−1.848	−2.651
	(2.573)	(1.999)
1.00–1.25	−0.264	−0.523
	(1.953)	(1.556)
1.25–1.50 (reference)	0	0
Predicted log residential land price	−0.004	−13.002***
	(4.029)	(3.073)
Observations	331,850	331,850
Panel C: Recorded theft		
0–0.25	−8.131**	−8.174***
	(3.320)	(2.546)
0.25–0.50	−4.133	−4.653*
	(3.239)	(2.476)
0.50–0.75	−3.523	−3.975*
	(3.126)	(2.386)
0.75–1.00	−2.254	−3.028
	(2.829)	(2.169)
1.00–1.25	0.050	−0.539
	(2.163)	(1.697)
1.25–1.50 (reference)	0	0
Theft rate (per 1,000 residents)	−0.008	0.040
	(0.040)	(0.034)
Observations	237,697	237,697
Nearest-facility fixed effects	Yes	Yes
Parent-500m fixed effects	Yes	Yes
Controls	No	Yes

Notes: The outcome is 2020 CWR; each band is compared with 1.25–1.50km. Panel A additionally requires positive 2015 child-cohort counts and all four sorting measures. Panel B uses the main sample; Panel C requires at least 50 residents. Each panel uses a common sample and women ages 15–49 weights. Migration rates are expressed in percentage points, land price is predicted log price, and theft is incidents per 1,000 residents. All 2020 disability models use disability service facilities, include facility-containing cells, and require positive counts of women ages 15–49 without a minimum count. The panel-specific migration, land-price, or theft regressors enter both columns. Both columns include nearest-facility and parent-500m fixed effects. Column (1) has no baseline controls; column (2) adds all 15 baseline controls. Standard errors in parentheses cluster by nearest facility and parent-500m mesh. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

that the number of children aged 0–4 changes because of two factors: the fertility rates and the migration of children.[23] My aim is to isolate the change due to the migration channel, but I cannot directly observe it in the Population Census because it is conducted every five years and the available migration variable is at the municipality level, which is a far more aggregate spatial unit than the 250m cell I use in this paper. Therefore, to proxy for children's net migration tendency, I follow changes in the numbers of children in two cohorts between the 2015 and 2020 Censuses: the cohorts with ages 5–9 and 10–14 in 2020. Since these changes do not reflect fertility decisions because these cohorts were already born, they reflect only the migration tendency. I call them "net migration rates of children." These measures proxy migration relevant to the 2020 CWR insofar as older and younger children's migration shares common determinants, implying that controlling for them would allow me to extract the impact of the disability service facilities on the fertility decisions.

Panel A of Table 3 includes both child-migration rates. The net migration rate for children ages 5–9 in 2020 enters the CWR regressions with the expected positive sign and is statistically significant at the 1 percent level, both without baseline controls (coefficient 0.077) and with all 15 baseline controls (coefficient 0.042). This positive association is consistent with shared migration patterns among families with older and younger children, providing indirect support for using the 5–9 measure as a proxy for migration among children ages 0–4.

The nearest-band coefficients are −6.771 without the baseline controls ($p = 0.039$) and −7.861 with the baseline controls ($p = 0.002$). Conditioning on these measured migration proxies attenuates the association in both specifications while leaving much of the negative CWR that is still statistically significant.

To further assess the role of migration, I additionally control for the log change in the number of women ages 15–49 and the female recent-mover share, defined in Appendix Equations (A.4) and (A.5). Changes in the female population might matter because facility proximity could influence where women of childbearing age live, thereby affecting the denominator of the CWR. I include the female recent-mover share because residential moves and childbearing decisions are interrelated so that migration may occur reflecting future fertility intentions (Kulu and Steele, 2013; Ermisch and Steele, 2016; Anstreicher and Venator, 2026). In Appendix Table B.10, including them alongside both child migration rates leaves a preferred nearest-band coefficient of −7.380 ($p = 0.002$). The coefficients remain negative in both specifications. Taken together, controlling for various migration-related factors does not explain a large part of the lower fertility rate around the disability service facilities.

Residential land prices. Housing markets can affect fertility through both the cost of accommodating children and households' financial resources (Lovenheim and Mumford, 2013; Dettling

[23] Precisely speaking, it is also affected by the infant mortality rate. I ignore this in this paper because child mortality in Japan is very low: the 2020 life tables imply survival above 99.7 percent from birth to age 5 and above 99.9 percent from age 5 to age 15 for both sexes (Ministry of Health, Labour and Welfare, 2021). Therefore, its spatial variation quantitatively cannot explain the variation in the CWR found in Section 4.

and Kearney, 2014). At the same time, proximity to disability service facilities could be associated with local prices through changes in housing demand. Indeed, although the evidence is mixed, the literature on estimating the impact of group homes on housing prices suggests that such an impact could exist (Gabriel and Wolch, 1984; Farber, 1986; Colwell et al., 2000; Galster et al., 2004). If so, the disability service facilities could affect the fertility rate through the housing market, not through the incidental social contacts.

I construct land price estimates for each 250m cell to assess this possibility. Specifically, for each centroid of the cell, I pick the two nearest residential-use assessment points in the 2020 Land Price Publication and Prefectural Land Price Survey. I then construct the predicted land price by weighting log prices by inverse squared distance; Appendix A.9 gives the details.

Panel B of Table 3 reports the results. Without baseline controls, controlling for land price hardly changes the nearest-band coefficient, from −6.711 to −6.710. With all 15 controls, it moves from −7.757 to −7.330: the absolute magnitude falls by 5.5 percent, and the CWR association remains negative and statistically significant ($p = 0.002$). Overall, facilities may raise nearby land prices, but controlling for land prices leaves the negative CWR association largely unchanged.[24]

Safety level. An alternative explanation for the lower fertility rate around the disability service facilities is that concerns about children's safety discourage childbearing. Prospective parents who expect a child to face violence or other dangers in the neighborhood may postpone or forgo a birth, although the effect could be the opposite once general-equilibrium effects are taken into account (Awaworyi Churchill et al., 2022; Guerra-Cújar et al., 2024). Child-safety concerns also appear in opposition to disability facilities in Japan. In a documented campaign in Yokohama, opponents of a disability group home displayed banners urging residents to "protect children's safety" (Okahara et al., 2022, p. 44).

While this could potentially explain the lower fertility rate around disability service facilities, official crime statistics in Japan do not show a strong association between crime rates and the disability service facilities. I use two complementary datasets to check this. First, I use the nation-level theft occurrence data. While this covers all of Japan and the spatial unit is granular, the data availability is limited only to the thefts. Second, I use the crime data of Tokyo, which have similar spatial granularity but cover not only thefts but other crimes, such as murders and burglary. To summarize, I do not find evidence that the safety level is worse near disability service facilities. In the nationwide theft data, I find that the nearest-band coefficient is statistically indistinguishable from zero. In the Tokyo model with facility and parent-1km fixed effects and its six baseline controls, violent or serious offenses and burglary are lower in the nearest band, while total crime and non-burglary theft are not significantly different. See Appendices B.8.1 and B.8.2 for more details.

[24]When predicted log land price is the outcome (Appendix Table B.11), the nearest-band coefficients are 0.0432 and 0.0329 in columns (1)–(2); the latter corresponds to approximately 3.34 percent higher predicted prices. This positive association is consistent with Farber (1986) and Galster et al. (2004).

As another approach to assess how much safety consideration could change my results, Panel C of Table 3 adds the 2020 recorded theft rate per 1,000 residents to each of the two CWR specifications.[25] In the specification of column (2) with baseline controls, adding the theft rate hardly changes the 0–250m coefficient, from −8.178 to −8.174 ($p = 0.001$). Overall, although I cannot fully rule out the possibility that crimes and other incidents not appearing in the official crime records around the disability service facility affect the fertility rates, they find essentially no support in the official crime statistics.

Sensitivity to unobserved confounding. In order to further mitigate the concern that the unobserved confounders drive my results, Appendix Table B.14 applies the proportional-selection analysis of Oster (2019), using migration, land price, and recorded theft as separate observed benchmarks and as a joint block. For the specifications with baseline controls, assuming equal proportional selection and an unobserved conditional R^2 gain equal to the observed gain yields adjusted coefficients of −7.300 and −6.338 for the two migration specifications, −6.902 for land price, and −8.169 for theft. Even in a highly conservative specification, Oster's lower bound is −5.047. Overall, my sensitivity analysis shows the robustness of the negative CWR around disability service facilities.

5 Additional analyses

This section examines additional heterogeneity and checks robustness to the measure of facility exposure, the sample definition, the fertility measure, and the method of estimation and inference.

Number of nearby facilities. My main analysis has focused on the distance from the nearest facility, but discarded information about how many disability service facilities exist. This could matter because it may affect the intensity of the incidental encounters. To analyze this, I replace the distance-band indicators with indicators for one, two, three, and four or more disability facilities within 250m of the cell's centroid, relative to zero facilities (Appendix Table B.15). With the same main sample and all 15 controls, the coefficients are −2.87, −4.34, −5.84, and −8.42. Their absolute magnitudes increase monotonically, indicating a stronger negative association with CWR as facility counts rise. This is consistent with the idea that the lower CWR is induced by more intensive incidental encounters.

Heterogeneity by female age and education. Here, I examine whether exposure to disability service facilities has heterogeneous impacts by women's age and education status. For this purpose, I allow all five distance-band coefficients to vary with the cell-level mean age of women ages 15–49 and the local university-graduate share, retaining all 15 baseline controls (Appendix

[25] I chose the theft variable because it is available nationwide.

Table B.3). I also allowed for the heterogeneity by private-car use as in Section 4.2.2 to account for the possibility that women's age and education might be correlated with the car usage. I find that the lower CWR around disability service facilities is more salient in a cell with younger women and a lower local university-graduate share. Separate joint tests of the five interactions for each attribute yield $p < 0.001$ for age and $p = 0.033$ for education. One possible interpretation of the age pattern is that women at earlier stages of family formation have relatively little information about the disability risk and the care needs or they may have an option of postponing childbearing decisions. The education pattern could reflect differences in prior information about disability and available support, or in resources for accommodating anticipated caregiving demands.

Facility-containing cells. The main analysis includes cells containing disability service facilities. Table B.16 instead excludes these cells and reconstructs the fixed-effect support. The preferred 0–250m coefficient is -7.001 ($p = 0.004$, $N = 294,946$), compared with -7.76 in the main sample.

Alternative CWR denominator. My main definition of the CWR used the ages between 15 and 49 as the childbearing age. Table B.17 instead uses women ages 20–39 in the denominator, which focuses more on the prime age of childbearing. Repeating the same analysis as Section 4, the nearest-band coefficient is -13.140 ($p = 0.004$), compared with -7.677 using women ages 15–49. The magnitudes are not directly comparable because the denominator changes, but the fully controlled association remains negative and statistically significant.

Two-kilometer estimation window. As seen in Figure 1, I restricted the sample to cells within 1.5km from the disability service facility. Table B.18 extends the sample to 2km while retaining 1.25–1.50km as the reference band. The preferred nearest-band coefficient is -7.714, close to the main estimate. Neither of the two added outer-band coefficients is statistically significant, suggesting that the main 1.50km sample encompasses sufficient spatial scope.

Spatial inference. While I have accommodated spatial correlation of standard errors by clustering at the nearest facility and the parent 500m mesh level, it does not accommodate spatial autocorrelation across different clusters. Table B.19 allows such spatial autocorrelation using Conley standard errors (Conley, 1999). Using this method does not change the point estimates. With distance cutoffs of 0.5, 1, and 2km, the nearest-band coefficient in the preferred specification remains significant at the corresponding p-values 0.0003, 0.0007, and 0.0010.

6 Conclusion

This paper sheds new light on a novel aspect of fertility decisions: the perceived probability and cost of having a child with a disability. Survey evidence from Japan and elsewhere suggests that

concerns about a child's health or disability enter stated childbearing decisions and are often comparable in importance to work-related concerns. Illustrative online accounts connect these concerns to everyday encounters with people with disabilities and their caregivers: individuals describe how seeing them in public spaces brings anticipated care needs to mind and contributes to their hesitation about having children.

Motivated by this evidence, I investigate whether fertility is lower near disability service facilities, which may provide more opportunities for incidental social contact with facility users and their caregivers. Using spatially granular census data in Japan, I find a lower child–woman ratio within 250m of the nearest disability service facility. Interpreted causally, the nearest-band estimate implies a fertility shortfall equivalent to approximately 1 percent of Japan's births. Including possible spillovers in the 250–500m and 500–750m bands raises this figure to approximately 1.8 percent. Conditional on facility distance, the CWR is also lower along walking routes to railway stations, and this association weakens as private-car use rises. Proximity differences are suggestively larger where car use is low. These patterns are consistent with the role of greater opportunities for incidental encounters. Placebos based on future disability facilities and elderly care facilities do not reproduce the main negative profile. Adjustments for selective migration, land prices, and recorded theft leave most of the association intact, further reinforcing the importance of incidental social contacts.

Together, my findings suggest that perceived disability risk and anticipated caregiving demands may influence childbearing and its impact could be large enough to affect a national fertility rate. While my results point to the role of incidental social contacts, the importance of other determinants of the anticipated disability risk or caregiving needs, such as education, mass media, friendship networks, and social media, remains for future research. At the same time, my result also suggests that the welfare system to support families with disabled children may have an ex ante insurance value as well as benefits for families already providing care. Confidence that financial assistance and care services will be available could ease uncertainty about parenthood, including for people who never use those services, and it might result in a higher fertility rate.

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Appendix to "Possible Futures in Sight: Disability Service Facilities and Local Fertility" (Not for Publication)

A Further details on the data **A2**

A.1 Census source files and harmonization . . . A2
A.2 Calculation of the national fertility shortfall . . . A2
A.3 Facility definitions and spatial sample . . . A3
A.4 Summary statistics . . . A5
A.5 Baseline controls and private-car use . . . A6
A.6 Transit data and walking-network construction . . . A6
A.7 New-entrant facilities and the 2015 analyses . . . A8
A.8 Residential sorting and migration measures . . . A8
A.9 Residential land-price measures . . . A9
A.10 Recorded-crime sources and geographic matching . . . A9
A.11 Alternative CWR denominator . . . A10
A.12 Sources for the motivating evidence in Section 2 . . . A10
A.12.1 Survey sources and question details . . . A10
A.12.2 Sources for the qualitative online accounts . . . A14

B Supplementary analyses **A16**

B.1 Facility overlap, population density, and regression support . . . A16
B.2 Transit-route direction and alternative paths . . . A18
B.3 Private-car-use heterogeneity . . . A20
B.4 Joint heterogeneity by private-car use, female age, and education . . . A23
B.5 Placebo analyses and 2015 cross-sectional replication . . . A24
B.5.1 Future entry of disability facilities placebo . . . A24
B.5.2 2020 CWR around the candidate future facilities . . . A26
B.5.3 2015 cross-sectional replication . . . A27
B.5.4 Elderly care facility placebo . . . A28
B.6 Residential sorting . . . A30
B.7 Residential land prices and neighborhood valuation . . . A31
B.8 Recorded crime as an alternative mechanism . . . A32
B.8.1 Nationwide recorded theft . . . A32
B.8.2 Tokyo town/chome crime . . . A33
B.9 Sensitivity to unobserved confounding . . . A34
B.10 Number of nearby facilities . . . A36
B.11 Robustness checks . . . A37

References for the Appendix **A41**

A Further details on the data

This appendix provides data sources and variable-construction details for the main and supplementary analyses.

A.1 Census source files and harmonization

The 2020 Population Census used for the main analysis follows the Statistics Bureau of Japan's regional-mesh system (Statistics Bureau of Japan, 2026; Statistics Bureau of Japan, 2025), which reports population by sex and five-year age group at the 250m mesh level. I download all 47 prefectural ZIP archives from e-Stat's Statistical GIS, sum prefectural component counts for mesh codes crossing prefectural borders, and retain a mesh only when every component of the age-population table is not subject to confidentiality restriction. This rule excludes both suppressed cells and cells receiving their aggregated counts.

The 2020 Census table also supplies education and residence-duration variables at 250m resolution (Statistics Bureau of Japan, 2022a). The university-graduate share is merged to the age table by mesh code. Housing counts also come from the 2020 Census 250m table (Statistics Bureau of Japan, 2020b). The private-car-use measure also comes from the census data (Statistics Bureau of Japan (2022b)). Construction of the housing and private-car-use shares is described in Appendix A.5.

The 2015 cross-sectional replication and future-entry placebo use the 250m mesh data from the 2015 Population Census (Statistics Bureau of Japan, 2015a; Statistics Bureau of Japan, 2015b). A comparable university-graduate share is unavailable, so the 2015 regressions omit it as a control variable.

To obtain changes in the female population, I merge the 2020 mesh data to the 2015 mesh data by identical mesh codes (Statistics Bureau of Japan, 2025; Statistics Bureau of Japan, 2015a).[A.1] The child cohort measures in Appendix A.8 also require merging the 2015 and 2020 mesh data (Statistics Bureau of Japan, 2015a; Statistics Bureau of Japan, 2020a). The measure for ages 5–9 in 2020 pairs the 2015 population ages 0–4 with the 2020 population ages 5–9; the measure for ages 10–14 in 2020 pairs ages 5–9 in 2015 with ages 10–14 in 2020. The female recent-mover share uses the sex-specific residence-duration items in the 2020 data.

A.2 Calculation of the national fertility shortfall

This section explains the national fertility shortfall reported in Section 4.1. Because the CWR is the number of children ages 0–4 per 1,000 women ages 15–49, I divide each estimated CWR reduction by 1,000 to express it as fewer children per woman. Multiplying this by the number of

[A.1] For spatial assignment in the 2015 analyses, JGD2000 centroids and the original four-corner cell footprints are transformed to JGD2011 using the official PatchJGD correction; unsupported coordinates are excluded. The largest applied displacement among clean cells with positive women counts is 5.65m. The correction leaves the matching of population counts unchanged.

women in the distance band gives the estimated shortfall in children in that band. I then compare the total with Japan's 4,516,082 children ages 0–4 in the 2020 Population Census (Statistics Bureau of Japan, 2021, p. 55).

Let R be the set of distance bands included in the calculation, W_b the number of women ages 15–49 in band b of the estimation sample, and $\widehat{\beta}_b$ the corresponding coefficient in column (2) of Table 2. The estimated shortfall in children ages 0–4 and its share of the national total, $C_J = 4{,}516{,}082$, are

$$\widehat{\Delta C}_R = -\frac{1}{1000}\sum_{b\in R} W_b\widehat{\beta}_b, \qquad \widehat{r}_R = \frac{\widehat{\Delta C}_R}{C_J}. \tag{A.1}$$

All calculations use the same regression estimates and sample, with 1.25–1.50km as the reference band. Interpreting the CWR differences causally, this shortfall approximates the number of births forgone over October 2015–September 2020, abstracting from child mortality and migration.

The 0–250m estimate gives a shortfall of 47,777 children, or 1.06 percent of the national total (95 percent confidence interval: 0.42–1.69 percent). Including possible spillovers in the 250–500m and 500–750m bands raises the shortfall to 83,014 children, or 1.84 percent (0.35–3.33 percent). These are the approximately 1 percent and 1.8 percent figures in the main text. Dividing by five gives approximately 9,600 and 16,600 births per year. Including all five distance bands gives 1.93 percent (0.31–3.56 percent), or approximately 17,500 births per year.

To express these magnitudes in terms of the total fertility rate (TFR), I use the mean of the national TFRs in 2016–2020: 1.44, 1.43, 1.42, 1.36, and 1.33 (Ministry of Health, Labour and Welfare, 2022). Multiplying their mean, 1.396, by $\widehat{r}_R$ gives approximately 0.015 TFR points for 0–250m and 0.026 points for 0–750m, assuming the same proportional change in fertility at every maternal age.

A.3 Facility definitions and spatial sample

Facility locations come from the 2021 National Land Numerical Information Welfare Facilities Data (Ministry of Land, Infrastructure, Transport and Tourism, 2021). I retain point coordinates, facility names, classification codes, and positional accuracy. After removing duplicates defined by location, municipality code, name, and minor classification, the file contains 266,572 points. To secure positional accuracy, I retain observations with positional accuracy codes 1 and 2.

I define disability service facilities using major code 03, disability support facilities and related establishments (障害者支援施設等); major code 04, facilities supporting the social participation of people with physical disabilities (身体障害者社会参加支援施設); and middle codes 9911, disability service facilities (障害者サービス施設), 0508, residential facilities for children with disabilities (障害児入所施設), and 0509, child development support centers (児童発達支援センター).[A.2] This definition yields facility records at 50,583 unique physical

[A.2] Official classification lists: major codes, middle codes, and minor codes.

locations. Special-needs schools (特別支援学校) are excluded from the main facility definition because they are educational institutions under the School Education Act, rather than welfare service facilities.[A.3] I verify that including them does not change my conclusion at the end of this section.

The elderly care facility placebo includes all records in major code 02, welfare facilities for older people (老人福祉施設), or middle codes 9907, fee-based homes for older people (有料老人ホーム); 9908, recreation facilities for older people (老人憩の家); and 9910, long-term-care insurance and related facilities (介護保険等の施設). Code 02 covers residential homes, welfare and day-service centers, short-stay facilities, home-care support, and supported living, including other and unspecified subcategories. Code 9907 includes service-equipped housing classified as a fee-based home. Code 9910 covers nursing homes, long-term-care welfare and health facilities, medical and integrated care institutions, and other care services. I retain elderly facilities strictly more than 1km from the closest admissible disability service facility in the complete catalog, leaving 22,704 records at 15,492 locations. The 1km restriction applies only to facility locations. Cells are reassigned to the nearest retained elderly facility.

For each facility definition, I retain cells whose centroids are strictly less than 1.5km from their assigned facility. Columns (1), (2), and (4) of Table 2 use this common main sample; columns (3) and (5) use the common route sample documented below. In each sample, I iteratively remove cells in singleton facility or parent-500m groups until no singletons remain. Where both specifications are reported, they use the same sample selected for the fully controlled model and include both fixed effects. The final disability sample contains 331,850 cells, including 27,493 facility-containing cells and 90,162 cells with fewer than ten women; its facility and parent-500m group counts are 36,377 and 101,755. The elderly distance-only sample contains 176,077 cells and 11,951 facility groups.

The data provide limited detail on service types. Of the 36,377 nearest-facility groups in the main 331,850-cell sample, 31,750 (87.3 percent) fall under category 9911, disability service facilities (障害者サービス施設). Its only minor code, 991100, does not distinguish specific services or residential and nonresidential facilities. These counts refer to the facility record assigned to each cell, rather than all records at the same location. The broad classification and small numbers of facilities in some remaining categories limit comparisons by service type.

As a robustness check, I add special-needs schools (classification code 16012) from the 2021 National Land Numerical Information Schools Data.[A.4] I reassign each cell to its nearest disability service facility or school and apply the same sample-selection rules within 1.5km. The regression retains all 15 controls, weights cells by women ages 15–49, and uses nearest-facility-or-school and parent-500m fixed effects with two-way clustering by these groups. Estimating a common distance profile for disability service facilities and schools gives 338,932 cells and a 0–250m coefficient of −6.85 (SE 2.34, $p = 0.003$), relative to 1.25–1.50km. This is close

[A.3] Article 1 of the School Education Act includes special-needs schools among the institutions defined as schools.

[A.4] Ministry of Land, Infrastructure, Transport and Tourism, National Land Numerical Information: Schools Data (P29), 2021 edition, accessed October 10, 2026.

to and statistically indistinguishable from −7.76 in the main analysis: the CWR remains lower near facilities under the broader definition.

A.4 Summary statistics

Table A.1 reports statistics weighted by women ages 15–49 in the main estimation sample.

Table A.1: Summary Statistics for the Main Estimation Sample

Variable	Mean	SD	p25	p50	p75	*N*
A. Outcome, exposure, and population						
Child–woman ratio	187.66	102.78	123.60	172.25	232.56	331,850
Distance to nearest facility (km)	0.426	0.311	0.198	0.340	0.568	331,850
Total population	687.04	550.95	298	540	928	331,850
Women ages 15–49	152.48	140.62	58	111	201	331,850
Children ages 0–4	27.91	32.42	9	20	36	331,850
B. Controls						
ln(population density)	8.920	0.905	8.424	9.016	9.556	331,850
Women ages 20–24 / women 15–49 (%)	12.13	6.44	8.37	11.57	15.00	331,850
Women ages 25–29 / women 15–49 (%)	12.25	6.07	8.33	11.88	15.70	331,850
Women ages 30–34 / women 15–49 (%)	13.13	5.91	9.52	12.90	16.39	331,850
Women ages 35–39 / women 15–49 (%)	14.84	5.97	11.46	14.49	17.69	331,850
Women ages 40–44 / women 15–49 (%)	16.82	6.12	13.33	16.36	19.85	331,850
Women ages 45–49 / women 15–49 (%)	19.67	6.78	15.62	19.35	23.40	331,850
University-graduate share (%)	25.33	11.34	16.83	23.96	32.77	331,850
Residential zoning indicator	0.646	0.478	0	1	1	331,850
Commercial zoning indicator	0.097	0.296	0	0	0	331,850
Industrial zoning indicator	0.088	0.284	0	0	0	331,850
Row-house share (%)	1.46	3.88	0.00	0.25	1.46	331,850
Apartment share (%)	47.97	30.73	21.53	48.44	74.66	331,850
Other/unstated housing share (%)	0.10	0.58	0.00	0.00	0.00	331,850
ln(station access distance in km)	0.192	0.806	-0.317	0.178	0.690	331,850

Notes: Statistics describe the main estimation sample and are weighted by women ages 15–49. CWR is children ages 0–4 per 1,000 women ages 15–49. The omitted categories are women ages 15–19 and detached houses.

A.5 Baseline controls and private-car use

Population density is total population divided by the spherical area of the 250m cell and enters the baseline vector as ln(density). Female age composition is represented by six shares: women ages 20–24, 25–29, 30–34, 35–39, 40–44, and 45–49, each divided by women ages 15–49. Ages 15–19 form the omitted category.

The university-graduate share divides university and graduate-school graduates ages 15 or older by all final-school graduates ages 15 or older, pooling men and women. I use the published graduate total without subtracting cases with an unreported final school. Housing shares use all general households living in housing as the denominator. Row-house and apartment counts come directly from the table; other or unstated forms are the residual after subtracting detached houses, row houses, and apartments. Detached houses form the omitted category.

Zoning is constructed from the 2019 National Land Numerical Information (Ministry of Land, Infrastructure, Transport and Tourism, 2019). Each 250m cell centroid is spatially joined to the zoning GIS data. Zoning codes 1–7 and 21 are classified as residential, 8–9 as commercial, and 10–12 as industrial. All three indicators equal zero for other designated zones and for cells without zoning, which together form the omitted category.

The baseline vector contains 15 controls: log population density, six female age shares, the university-graduate share, three zoning indicators, three housing shares, and log station-access distance, whose construction is described in the next section. Table 2 omits this vector in column (1) and includes all 15 controls in columns (2)–(5); every column includes both fixed effects. Supplementary analyses generally report two specifications on the same sample: (1) both fixed effects without baseline controls, and (2) both fixed effects with all applicable baseline controls. The 2015 models omit the university-graduate share because the corresponding values in 2015 are not available, leaving 14 controls.

The private-car-use measure is constructed from the 2020 Census 250m table. I aggregate the published counts from all constituent 250m cells to each parent 500m mesh. The rate divides private-car users commuting to work or school by resident workers and students excluding people working at home. I assign the parent rate to each constituent cell. Respondents report every commuting mode used, so the rate measures any private-car use rather than car-only commuting (Statistics Bureau of Japan, 2020c).

A.6 Transit data and walking-network construction

Using the 2020 railway data (Ministry of Land, Infrastructure, Transport and Tourism, 2020c), I group station features representing the same station and retain a fixed catalog of 9,088 station groups. Each station's representative point is a medoid of the group's distinct feature coordinates. I also use the 2022 bus-stop data (Ministry of Land, Infrastructure, Transport and Tourism, 2022), which yield 242,984 unique coordinate points.

Walking paths are traced on the January 1, 2021 historical OpenStreetMap network for Japan, supplied in eight Geofabrik regional extracts (OpenStreetMap contributors, 2021), using GraphHopper 11.0 (GraphHopper, 2026). The nationwide pedestrian network excludes ferry edges and minimizes road distance. For a facility, I compute the walking distance to the five closest station groups or bus-stop points in great-circle distance. Among successfully routed candidates, I select as the "distance to the nearest train station" the smallest sum of network distance and straight connectors at both endpoints, breaking ties by straight-line candidate rank. Quality restrictions apply to the selected route, without replacement by another candidate. The distance to the nearest bus stop is constructed in the same way.

For placebo paths, I rotate the selected actual destination around the facility by 90, 180, or 270 degrees in local east–north coordinates and recompute the walking route to each rotated destination. This preserves the facility anchor and straight-line endpoint distance, while routed lengths can change. For each direction and destination type, a cell's indicator is one if its rectangle intersects the available walking-network route or either endpoint connector assigned to its nearest facility, zero if the path is available but neither intersects, and missing if the path is unavailable.

The facility and destination coordinates may lie off the mapped walking network, especially when the destination is a rotated placebo endpoint. GraphHopper connects each coordinate to the nearest road segment accessible to pedestrians. I retain a facility path only if routing succeeds and each endpoint's straight connector is no greater than 1,000m.

The station-path specification used in Section 4.2.1 and the supplementary bus-path specification, reported in Appendix Tables B.1 and B.2, require all four station paths and all four bus paths to be available. This leads to 315,357 cells assigned to 34,457 facility groups. Each regression includes only its own destination type's four indicators, together with the five distance-band indicators. Among these facility groups, median actual network distances are 1.399km to the selected station and 0.252km to the selected bus stop; including endpoint connectors, the corresponding medians are 1.442km and 0.275km, and 44.9 percent of bus paths are shorter than 250m. The elderly station-path regression uses the same route-availability requirements and contains 163,197 cells in 10,648 facility groups.

While the above procedure yields the facility-to-station path, the cell-to-station path is constructed separately to define the distance to the nearest station for each cell. From each cell centroid, I consider the five station groups closest in great-circle distance and select the shortest successfully computed route on the same non-ferry network, including straight-line connectors at both endpoints. I screen the selected route for unusually large detours and excessive reliance on endpoint connectors. These restrictions reduce sensitivity to gaps in the mapped network and off-network connections whose walkability is unverified.[A.5] I retain only positive, finite total distances no greater than 100km and use $\ln(L/1{,}000)$ as the regressor, with L measured in

[A.5] Let L denote total route length, S the straight-line distance to the selected station, and a and b the endpoint connector lengths, all in meters. Define $C = a + b$. I exclude the selected route if (i) $L \geq 5{,}000$ and $L \geq 5S$; (ii) $S \leq 250$, $L \geq 1{,}000$, and $L \geq 5S$; (iii) either connector exceeds 250m; or (iv) $C > 100$ and $C \geq 0.25L$.

meters. Successful routing to all five candidates is not required. Quality restrictions are applied after route selection, without switching to another station if the selected route fails the checks.

A.7 New-entrant facilities and the 2015 analyses

The future-entry placebo screens individual disability service facility records observed in 2021 against the two older (2011 and 2015) versions of the same welfare-facility datasets (Ministry of Land, Infrastructure, Transport and Tourism, 2011; Ministry of Land, Infrastructure, Transport and Tourism, 2015; Ministry of Land, Infrastructure, Transport and Tourism, 2021). Because the 2015 facility dataset does not cover Osaka, I exclude both facility records and Census cells in Osaka. The resulting universe contains 56,481 accurately located disability service facility records.

For each 2021 facility, I check whether any welfare facility was recorded within 250m of its location in either 2011 or 2015. If so, I exclude it from the placebo candidates. The check covers all welfare-facility types, including elderly and child welfare facilities. I then compare each facility's name and address with the 2011 and 2015 records and exclude it if an earlier record in the same prefecture has the same name or address within 5km, or a very similar name within 2km. This procedure leaves facilities at 9,301 distinct locations.

The 2015 cross-sectional replication measures proximity to disability service facilities recorded in Ministry of Land, Infrastructure, Transport and Tourism (2015). The selection covers disability-service classifications and fine-classification codes explicitly identifying residential institutions for children with disabilities, child-development support centers, or named predecessor institutions. Generic child-welfare, consultation, after-school day-care, and ambiguous medical or service classifications are not included. Osaka is omitted because its 2015 facility dataset is unavailable.

A.8 Residential sorting and migration measures

I construct the four migration-related measures used in Section 4.3.2 as follows.

For children, let C_{it}^{a-b} denote the number of children ages a through b in cell i in year t. The net child migration rate (5–9 as of 2020) is defined as

$$S_i^{C,1} = 100\left(\frac{C_{i,2020}^{5\text{–}9}}{C_{i,2015}^{0\text{–}4}} - 1\right), \qquad C_{i,2015}^{0\text{–}4} > 0. \tag{A.2}$$

The net child migration rate (10–14 as of 2020) follows the next older cohort:

$$S_i^{C,2} = 100\left(\frac{C_{i,2020}^{10\text{–}14}}{C_{i,2015}^{5\text{–}9}} - 1\right), \qquad C_{i,2015}^{5\text{–}9} > 0. \tag{A.3}$$

A rate of zero indicates unchanged cohort population. Ignoring mortality, which is very low among children in Japan, positive and negative values proxy net inflows and outflows as a percentage of the 2015 cohort.

Let W_{it} denote the number of women ages 15–49 in cell i in Census year t. Their population change is defined as

$$G_i^W = 100 \left[\ln W_{i,2020} - \ln W_{i,2015} \right]. \tag{A.4}$$

Both counts must be strictly positive.

The female recent-mover share is

$$R_i^W = 100 \times \frac{F_{i,<1\text{ year}} + F_{i,1\text{–}5\text{ years}}}{F_{i,\text{birth}} + F_{i,<1\text{ year}} + F_{i,1\text{–}5\text{ years}} + F_{i,5\text{–}10\text{ years}} + F_{i,10\text{–}20\text{ years}} + F_{i,20+\text{ years}}}, \tag{A.5}$$

where the categories cover female residents of all ages in 2020 with known duration at their current address.[A.6] Residents with unknown duration are excluded from the denominator.

A.9 Residential land-price measures

The residential land-price measure pools the 2020 Land Price Publication, assessed on January 1, and Prefectural Land Price Survey (L02), assessed on July 1 (Ministry of Land, Infrastructure, Transport and Tourism, 2020a; Ministry of Land, Infrastructure, Transport and Tourism, 2020b). I retain points whose recorded current use is exclusively residential and predict the natural logarithm of land price in yen per square meter at each 250m cell centroid using the two nearest points, weighting their log prices by inverse squared distance. The unweighted median distance to the second point is 0.918km in the main sample. Appendix B.7 describes its use as a control and an outcome.

A.10 Recorded-crime sources and geographic matching

The nationwide theft database was compiled and geocoded by Mamoru Amemiya and collaborators from prefectural-police open data (Amemiya and Yamane, 2021; Amemiya and Tanaka, 2022). Its seven theft categories are purse snatching, theft from vehicles, vehicle-parts theft, vending-machine theft, motor-vehicle theft, motorcycle theft, and bicycle theft. The 2020 file contains 180,646 records and is distributed through the University of Tsukuba's Division of Policy and Planning Sciences Commons Data Bank.

The compilers match police-reported town/chome or oaza addresses to representative coordinates, rather than exact crime locations. Retaining records with valid coordinates and a reported neighborhood leaves 179,194 records, or 99.20 percent of the 2020 source file. Retained records are assigned to 250m meshes by their coordinates and counted. The theft rate

[A.6] Each F counts female residents in the indicated residence-duration category; $F_{i,\text{birth}}$ counts those living at the same address since birth.

divides these counts by 2020 cell population and multiplies by 1,000. Both nationwide theft-outcome regressions and CWR regressions controlling for theft require at least 50 residents per 250m cell to reduce noise. Recursive removal of facility and parent-500m singletons then yields the common 237,697-cell sample.

The second source is the Tokyo Metropolitan Police Department's annual town/chome crime table for 2019–2021 (Tokyo Metropolitan Police Department, 2026). Normalized place names are matched to 2020 Census small-area boundaries, populations, and representative points from e-Stat (Statistics Bureau of Japan, 2020d). These data include all recorded Penal Code offenses and allow separate outcomes for violent or serious offenses, burglary, and non-burglary theft. Tokyo crime rates use 2020 population and are expressed per 1,000 residents. To be in the sample, I require town/chome areas have at least 50 residents to reduce noise, and a representative point less than 1.5km from the nearest admissible disability service facility. Both town/chome specifications use population weights and nearest-facility and parent-1km fixed effects on the same complete-case, recursively non-singleton sample. As controls, I also construct the following: log population density, log area, three zoning indicators, and log station access. Both specifications use the same station-access eligibility rule.

A.11 Alternative CWR denominator

The alternative CWR retains the age-0–4 numerator in Equation (1) and restricts the denominator to women ages 20–39:

$$\mathrm{CWR}_i^{20\text{–}39} = 1000 \times \frac{\text{children ages 0–4}_i}{\text{women ages 20–39}_i}. \tag{A.6}$$

Table B.17 compares the two CWR definitions on a common sample with strictly positive counts of women ages 20–39 and the required fixed-effect support. Each regression uses its own female-population denominator as the analytic weight.

A.12 Sources for the motivating evidence in Section 2

This section documents the survey evidence in Section 2.1 and the qualitative online accounts in Section 2.2.

A.12.1 Survey sources and question details

Table 1 compares concerns about a child's health or disability with work-related reasons for not wanting children. Its two percentage columns use the same question, response format, and denominator within each survey and subgroup. Years refer to fieldwork. The JPSC and JILI rows report overall shares, with shares for women with one child or less in parentheses.

Nippon Foundation, 2024. Public source: report and tabulations, pp. 3 and 21–22.

The *Survey of Attitudes toward the Declining Birthrate* was conducted online on September 26–30, 2024, among 6,000 men and women aged 15–45 across Japan, with allocations by sex and age within six regions (Nippon Foundation, 2024, pp. 3, 21–22). Question Q06 asks childless respondents who do not want children or are content to remain childless for their reasons (multiple responses). Among these 1,580 respondents, 21.5 percent selected "子どもが健康に育つか不安だから" (concern about whether a child would grow up healthy). The item concerns health throughout childhood.

The published sex-by-age tabulations report child-health concern shares of 19.7, 21.3, and 17.3 percent among men aged 15–25, 26–35, and 36–45 (n = 239, 239, and 284), respectively; the corresponding shares among women are 20.7, 26.0, and 24.0 percent (n = 261, 219, and 338) (Nippon Foundation, 2024, p. 22).

The work-related reason reported in Table 1 is "仕事と育児の両立が難しいから" (difficulty combining work and childcare), selected by 19.4 percent of the same 1,580 respondents. A separate option, "自分のキャリアを優先したいから" (prioritizing one's career), was selected by 8.5 percent.

Japanese Panel Survey of Consumers, 2021. The JPSC microdata were provided by the Panel Data Research Center, Institute for Economic Studies, Keio University. Table 1 uses wave 29, fielded in 2021, from the 1993–2021 data collection (Institute for Research on Household Economics and Panel Data Research Center, Institute for Economic Studies, Keio University, 2024). The figures are my calculations from the microdata. The married questionnaire asks women of all ages whether they want a child or another child (Q511), and asks those answering no for their reasons (Q513, multiple responses). The health item, Q513I, is "丈夫な子どもが生まれるかどうか不安だから" (concern about whether a healthy child would be born); Q513C is "仕事を続けたいから" (wanting to continue working), and Q513A is "教育費・養育費の負担が大きいから" (the burden of education and child-rearing expenses).

The table includes all 596 married women aged 49 or younger in the 2021 survey who reported that they did not want to have a child or another child. No lower age cutoff is imposed; the observed age range is 32–49 because the youngest respondents in wave 29 are aged 32. Current child counts (Q57) are observed for all of them: 138 have zero or one child and 458 have two or more. Health concerns were selected by 85 women overall (14.3 percent): 30 women with zero or one child (21.7 percent) and 55 women with two or more children (12.0 percent). Wanting to continue working was selected as a reason by 63 women overall (10.6 percent): 12 women with zero or one child (8.7 percent) and 51 women with two or more children (11.1 percent). Education and child-rearing costs were selected by 42 women with zero or one child (30.4 percent) and 205 women with two or more children (44.8 percent). As in the JILI tabulation, denominators include eligible women who did not answer the reason questions.

JPSC pooled microdata analyses. The child-count analysis pools 5,487 person-year observations from 1,277 married women aged 24–39 in 2004–2021 who did not want a child or another child, with valid responses to all six reasons and complete covariates. Panel A of Table A.2 reports OLS regressions of current child counts on six jointly entered reason indicators: health, expenses, work, housing, husband's support, and prioritizing one's own life. Controls are single-year age, survey-wave, prefecture, education, and employment-status indicators, the spousal age gap and its square, and whether the couple lives together.

The birth analysis uses 6,563 person-year observations from 1,803 married women aged 24–39 who did not want a child or another child at baseline, with valid health-reason and next-year birth responses. Baseline years are 1994, 1997, 2000, and 2004–2020. The outcome is a birth reported in the next annual wave (Q67A; 155 births). Panel B of Table A.2 compares two linear probability models on the same sample: one includes only the health-reason indicator and a constant; the other adds child-count categories (0, 1, 2, 3 or more) and the covariates listed above, all measured at baseline. Both analyses are unweighted, with standard errors clustered at the individual level.

Japan Institute of Life Insurance, 2002. Public source: SSJDA 0306 questionnaire and tabulations, Q17 follow-up, items 14 and 16.

The *Survey on Married Women's Life Planning*, archived as SSJDA study 0306, was conducted by mail on December 13–23, 2002; the report was published in March 2003 (Japan Institute of Life Insurance, 2003). It surveyed 879 married women aged 20–49 within the Greater Tokyo 30-km area, drawn from a survey company's registered panel; divorced or widowed women and students were excluded. Q17's follow-up asks women who do not want a child or another child for their reasons (multiple responses). Among these 590 women, 13.4 percent selected item 16, "丈夫な子どもが産まれるかどうか不安だから" (concern about whether a healthy child would be born).

The work-related reason is item 14, "仕事ができなくなる可能性があるから" (the possibility of being unable to work), selected by 15.6 percent of the same 590 respondents.

The subgroup shares and regressions use microdata supplied by the Social Science Japan Data Archive, study 0306 (https://doi.org/10.34500/SSJDA.0306), without weights. The analysis includes the 590 women with Q17=0 and excludes the other 289 respondents, whose reasons are missing. Four eligible women with no reason response retain their recorded zeros, matching the published denominator. Q16 records current child counts, not completed fertility. The parenthetical figures in Table 1 pool 29 childless women and 93 women with one child ($n = 122$).

The financial and housing items are "教育費・養育費などの負担が大きそうだから" (expected education and child-rearing expenses) and "十分なスペースのある住宅に住めそうもないから" (difficulty securing sufficiently spacious housing). Education and child-rearing

expenses were cited by 48.4 percent of women with one child or less and 67.5 percent of women with two or more children; the corresponding housing shares were 19.7 and 38.9 percent.

The JILI estimates in Panel A of Table A.2 report the health, expense, and work coefficients from an OLS model of current child counts jointly including the six corresponding reason indicators listed above, single-year age and education indicators, and years since marriage. Six missing marriage-duration values leave 584 observations. These cross-sectional associations are descriptive: existing family size can itself shape reported reasons, so they do not identify effects on first or subsequent births. The health item is broader than disability, and the survey does not link concerns to facility exposure.

Table A.2: Stated Reasons, Existing Children, and Subsequent Births

Panel A: Current number of children, OLS coefficients	JILI (2002)	JPSC (2004–2021)
Concern about a healthy birth	−0.415***	−0.480***
	(0.099)	(0.065)
Education and child-rearing expenses	0.150*	0.053
	(0.089)	(0.041)
Work-related reason	−0.082	0.007
	(0.096)	(0.056)
Controls	Yes	Yes
Observations	584	5,487
Women	584	1,277
R^2	0.176	0.171

Panel B: JPSC, birth in the following year Coefficients and standard errors in percentage points	No controls	With controls
Concern about a healthy birth	−1.01**	−1.23**
	(0.49)	(0.50)
Person-year observations	6,563	6,563
Women	1,803	1,803
Births	155	155

Notes: Panel A jointly includes six reason indicators; three coefficients are shown. JPSC observations are person-years. Estimates are unweighted. Standard errors in parentheses are HC1 for JILI and clustered by individual for JPSC.

Panel A controls: JILI includes single-year age and education indicators and years since marriage; JPSC includes single-year age, education, survey-wave, prefecture, and employment-status indicators, the spousal age gap and its square, and spousal cohabitation. Panel B's controlled model includes the same JPSC controls plus child-count categories (0, 1, 2, 3 or more), all measured at baseline. See Appendix A.12.1 for sample definitions and interpretation. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

UNFPA/YouGov, 2024. Public sources: YouGov tabulations, pp. 13–14 and 16; UNFPA report, methodological note on p. 143.

The online survey used YouGov and partner panels in 14 countries between November 15 and December 5, 2024, targeting adults aged 18 or older. Table 1 reproduces the weighted percentages in the published tabulations (YouGov, 2025, pp. 13–14, 16). Full country samples, including parents, rated "Fear of having an ill or disabled child" and "Raising a child is too difficult to reconcile with paid work" as reasons for not wanting children now or previously. Each reason is rated separately on a four-point scale: very important, fairly important, not very important, and not important at all. In both percentage columns, the first percentage denotes very important and the second combines very and fairly important. The denominator includes the separate responses don't know and prefer not to say.

The country-specific sample sizes are 1,054 in the United States, 1,053 in Brazil, 1,066 in Germany, 1,039 in Hungary, 1,048 in India, 1,050 in Indonesia, 1,010 in Italy, 1,123 in Mexico, 1,117 in Morocco, 528 in Nigeria, 1,033 in South Africa, 1,057 in South Korea, 1,009 in Sweden, and 1,076 in Thailand. The survey covers the national adult population in most countries, Internet users in Indonesia, Morocco, Nigeria, and Thailand, and married adults in Morocco. The methodological note reports urban bias in Nigeria and South Africa, and to a lesser extent in Brazil, India, and Mexico.

A.12.2 Sources for the qualitative online accounts

Sources 1–3 correspond to the excerpts in Section 2.2; sources 4–5 correspond to the additional excerpts below.

1. GirlsChannel, comment 403, posted January 30, 2018 (accessed September 4, 2026).

2. GirlsChannel, comment 269, posted July 23, 2017 (accessed September 28, 2026).

3. Yahoo! Chiebukuro answer, posted March 18, 2026 (accessed September 4, 2026).

4. Mamastar, comment 276, posted January 26, 2017 (accessed September 4, 2026).

5. Mitsuko Tateishi,「自閉症の息子妊娠中、出生前診断を受けた私」["During Pregnancy with My Autistic Son, I Underwent Prenatal Testing"], *LITALICO Hattatsu Navi*, published January 17, 2022 (accessed October 3, 2026).

Additional online accounts.

(4) "I see people with Down syndrome at a disability facility […] I absolutely could not give birth after such a diagnosis."

(5) "My experience during the practicum at a special-needs school, where I saw children with disabilities, when I felt that I probably could not raise a child with a disability, was also part of the background to my decision to undergo prenatal testing."

Selection and interpretation. The purposive, non-systematic search focused on encounters outside the family. I excluded accounts in which the writer's own disability or a relative's disability was the principal source of concern. These selected examples illustrate individual perceptions. The translations retain the writers' language, including stigmatizing expressions, to convey those perceptions.

Complementary interview evidence. A related account comes from the U.S. interview-based doctoral dissertation by Kissling (2020, pp. 37–38). She links Alvin's decision to seek a vasectomy to his observation of the care required by children with disabilities whom his mother cared for. Alvin, a 29-year-old childfree man, explained, "I don't want to take that risk." The dissertation does not specify whether the children were related to Alvin.

B Supplementary analyses

Unless stated otherwise, regressions follow Equation (2): five 250m distance bands are compared with 1.25–1.50km, facility-containing cells are included, and weights are the number of women ages 15–49. Both specifications include nearest-facility and parent-500m fixed effects, with two-way clustering by these units. Column (1) has no baseline controls; column (2) includes all 15 baseline controls. See Appendix A for data sources and variable construction.

B.1 Facility overlap, population density, and regression support

Figure 1(b) illustrates the simplest case in which no 1.5km circle overlaps. Figure B.1 explains how I deal with the case of overlapping facility neighborhoods. Each cell enters once and is assigned to its nearest admissible disability service facility, so its distance is $d_i = \min_j d(i, j)$ where i denotes a cell and j denotes a disability service facility. That is, when a cell i is near several facilities, the closest facility determines its distance band. While this data construction implicitly disregards information about how many facilities each cell is exposed to, I consider this aspect in Section 5.

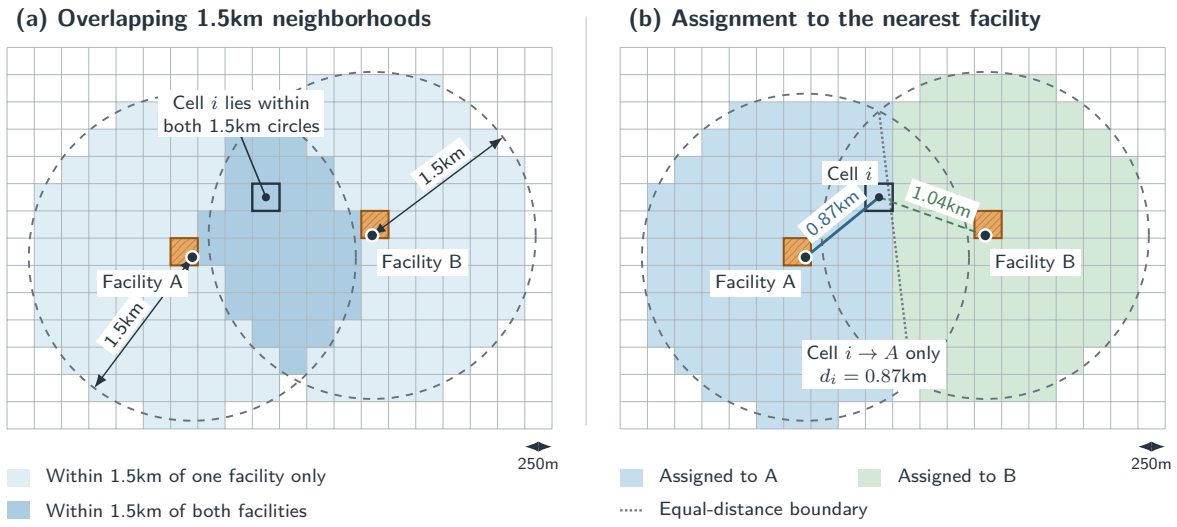


Figure B.1: Nearest-facility assignment with overlapping facility neighborhoods

Notes: Panel (a) shows overlapping 1.5km neighborhoods. Panel (b) assigns each cell to its nearest facility; the dotted line marks equal distances. Cell i is 1.04km from facility B and 0.87km from A. With B alone it enters the 1,000–1,250m band; with both facilities it enters the 750–1,000m band and facility A's fixed-effect group. Each cell is counted once, and its parent-500m fixed-effect group is unchanged. Orange hatched cells contain facilities and are included. Unshaded cells lie outside the 1.5km cutoff. Distances are illustrative; estimation uses great-circle distances from cell centroids.

Sample composition and regression variation. The above data construction implies that cells tend to be in a shorter distance band where facilities are closely spaced, as in urban areas. This raises a concern that the estimated difference in the distance band coefficients captures the different sample composition used for their estimation, rather than the true difference in the exposure to the disability service facility. To address this issue, Figure B.2 compares the population-density composition of each distance band with that of the variation used to estimate its coefficient. The latter is the variation in each distance-band indicator after accounting for the fixed effects, controls, and other distance bands. I measure each density group's share of this residualized variation using partial leverage (MacKinnon et al., 2023). Cells with at least 10,000 residents per km^2 account for 55.5 and 15.2 percent of women in the 0–250m and 1,000–1,250m bands, respectively, but 16.9 and 13.6 percent of the residualized variation for the corresponding coefficients. This variation has a broadly similar population-density composition across distance coefficients.

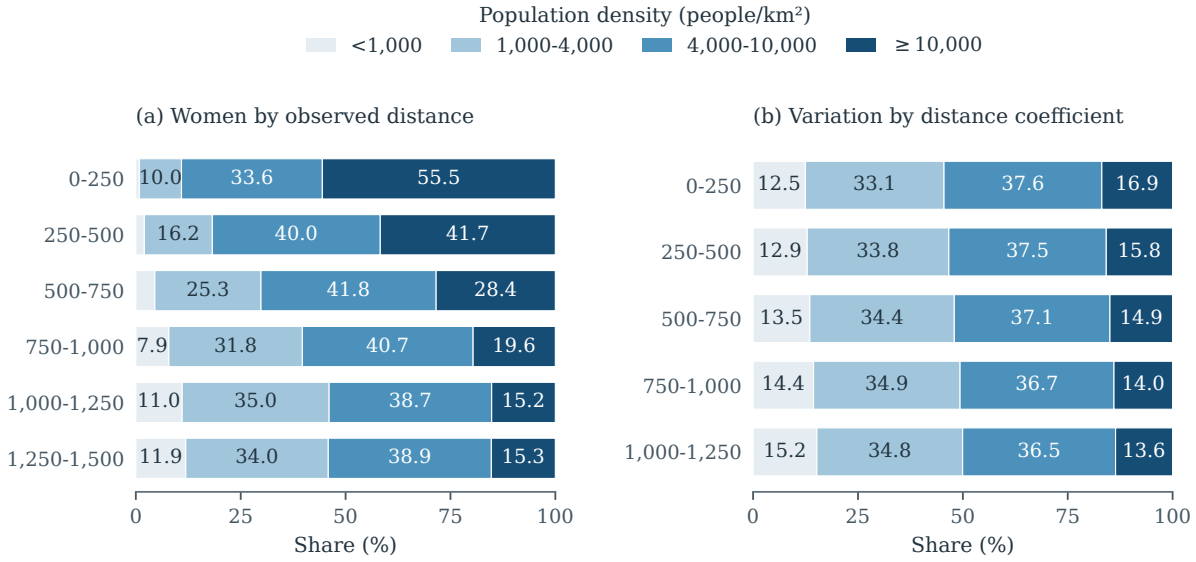


Figure B.2: Population density, sample composition, and regression support

Notes: Panel (a) shows the distribution of women ages 15–49 across population-density groups within each distance band. Panel (b) shows each group's share of the residualized distance-band variation in column (2) of Table 2, measured by partial leverage (MacKinnon et al., 2023). For each band b, let r_{ib} be the residual from a weighted regression of D_{ib} on nearest-facility and parent-500m fixed effects, all 15 controls, and the other four distance-band indicators, with w_i equal to women ages 15–49. Panel (b) sums $q_{ib} = w_i r_{ib}^2 / \sum_j w_j r_{jb}^2$ within each density group over all 331,850 sample cells, including the other distance bands and the 1.25–1.50km reference band.

B.2 Transit-route direction and alternative paths

The route analysis in Section 4.2.1 examines opportunities for incidental social contact by adding four path indicators to each of the two supplementary specifications: the actual station path and three rotated placebo paths. Appendix A.6 describes their construction.

Table B.1 reports the station-path specification used in the main text on 315,357 cells. Its fully controlled column (2) also appears in column (3) of Table 2 and panel (a) of Figure 4.

Table B.1: OSM Station Paths

Variable	(1)	(2)
0–0.25km	−4.099	−6.129**
	(3.202)	(2.466)
0.25–0.50km	−2.403	−3.758
	(3.046)	(2.344)
0.50–0.75km	−2.464	−3.372
	(2.922)	(2.248)
0.75–1.00km	−1.700	−2.543
	(2.630)	(2.028)
1.00–1.25km	−0.214	−0.442
	(1.995)	(1.576)
1.25–1.50km (reference)	0	0
Actual station path	−3.840***	−2.118***
	(0.856)	(0.686)
Endpoint rotated 90°	−0.920	−0.910
	(0.823)	(0.634)
Endpoint rotated 180°	1.480*	1.326**
	(0.832)	(0.635)
Endpoint rotated 270°	−1.133	−1.126*
	(0.819)	(0.631)
Facility fixed effects	Yes	Yes
Parent-500m fixed effects	Yes	Yes
Controls	No	Yes
Observations	315,357	315,357

Notes: The outcome is the 2020 CWR. Five distance bands (reference: 1.25–1.50km) and four station-path indicators enter jointly. Station and bus models share cells with all four station and four bus paths available. Column (1) has no controls; column (2) includes all 15. Models use women ages 15–49 as weights and two-way clustering by facility and parent-500m mesh (standard errors in parentheses). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Bus-stop robustness check. Table B.2 replaces the four station-path indicators in Equation (3) with bus-path indicators on the same 315,357 cells. The fully controlled actual bus-path coefficient is also negative (−2.459, $p = 0.002$), which is analogous to the main station-path result.

Table B.2: OSM Bus-Stop Paths

Variable	(1)	(2)
0–0.25km	−4.126	−6.003**
	(3.214)	(2.480)
0.25–0.50km	−2.666	−3.823
	(3.044)	(2.344)
0.50–0.75km	−2.650	−3.415
	(2.922)	(2.248)
0.75–1.00km	−1.800	−2.561
	(2.630)	(2.028)
1.00–1.25km	−0.261	−0.452
	(1.995)	(1.576)
1.25–1.50km (reference)	0	0
Actual bus-stop path	−2.518**	−2.459***
	(1.029)	(0.789)
Endpoint rotated 90°	−1.403	−0.417
	(0.976)	(0.755)
Endpoint rotated 180°	0.166	0.511
	(0.999)	(0.760)
Endpoint rotated 270°	−0.530	−0.371
	(1.020)	(0.777)
Facility fixed effects	Yes	Yes
Parent-500m fixed effects	Yes	Yes
Controls	No	Yes
Observations	315,357	315,357

Notes: The outcome is the 2020 CWR. Five distance bands (reference: 1.25–1.50km) and four bus-path indicators enter jointly. Station and bus models share cells with all four station and four bus paths available. Column (1) has no controls; column (2) includes all 15. Models use women ages 15–49 as weights and two-way clustering by facility and parent-500m mesh (standard errors in parentheses). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

B.3 Private-car-use heterogeneity

The model underlying Figure 5 allows each of the five distance-band coefficients to vary with private-car use.

Let $c_{m(i)}$ be the share of commuters and students using private cars in cell i's parent-500m mesh, expressed between zero and one. Define

$$z_{m(i)} = \frac{c_{m(i)} - \bar{c}_w}{0.10}, \qquad \bar{c}_w \simeq 0.41086,$$

where $\bar{c}_w$ is the mean car-use rate, weighted by the number of women at childbearing age. One unit change in $z_{m(i)}$ represents 10 percentage points. The indicator D_{ib} identifies distance band b, from $b = 0$ (0–250m) through $b = 4$ (1.00–1.25km), with 1.25–1.50km omitted. The specification is:

$$\begin{aligned} \mathrm{CWR}_i = & \sum_{b=0}^{4} \beta_b D_{ib} + \sum_{b=0}^{4} \theta_b z_{m(i)} D_{ib} \\ & + X_i'\gamma + \alpha_{f(i)} + \delta_{m(i)} + u_i. \end{aligned} \tag{B.1}$$

The coefficient θ_b measures the change per 10 percentage points of car use in band b relative to the omitted band. At car-use rate c, the CWR difference is $\beta_b + \theta_b(c - \bar{c}_w)/0.10$.

The fully controlled specification in column (4) of Table 2 uses 331,850 cells, nearest-facility and parent-500m fixed effects, and the 15 baseline controls. Parent-500m fixed effects absorb the car-use level, while the interactions remain estimable because distance can vary among constituent 250m cells.

Figure 5 evaluates this fully controlled specification at the women-weighted 10th, 50th, and 90th percentiles of car use (6.9, 37.6, and 79.2 percent). The 95 percent confidence intervals use the full clustered covariance matrix. Panel B of Table 2 reports the nearest-band conditional coefficients at these three percentiles in column (4), together with the joint test of the five distance-band interactions.

Transit-route heterogeneity. The specification that jointly considers distance from the disability facility and facility-to-train path for Section 4.2.1 is

$$\begin{aligned} \mathrm{CWR}_i = & \sum_{b=0}^{4} \big(\beta_b + \theta_b z_{m(i)}\big) D_{ib} + \sum_{q \in \{A,90,180,270\}} \big(\lambda_q + \kappa_q z_{m(i)}\big) R_i^q \\ & + X_i'\gamma + \alpha_{f(i)} + \delta_{m(i)} + u_i. \end{aligned} \tag{B.2}$$

Here $q = A$ denotes the actual station path and $q \in \{90, 180, 270\}$ the rotated placebo paths; θ_b and κ_q measure changes in the distance-band and path coefficients, respectively, per 10 percentage points of car use. The fully controlled joint model is reported in column (5) of Table 2 and uses the same 315,357 cells and estimation conventions as column (2) of Table B.1. The model without baseline controls omits $X_i'\gamma$ and retains all other terms on the same sample. Figure B.3

evaluates both joint models at the route-sample percentiles (6.8, 36.5, and 78.8 percent). Rows correspond to the 10th, 50th, and 90th percentiles; the left column omits baseline controls and the right column includes all 15. These estimates also appear in Panel B, column (5), of Table 2. The actual-path interaction is 0.924 per 10 percentage points of car use (SE 0.250, $p < 0.001$), and the four path interactions are jointly significant ($p < 0.001$). The negative station-path association thus weakens as private-car use rises.

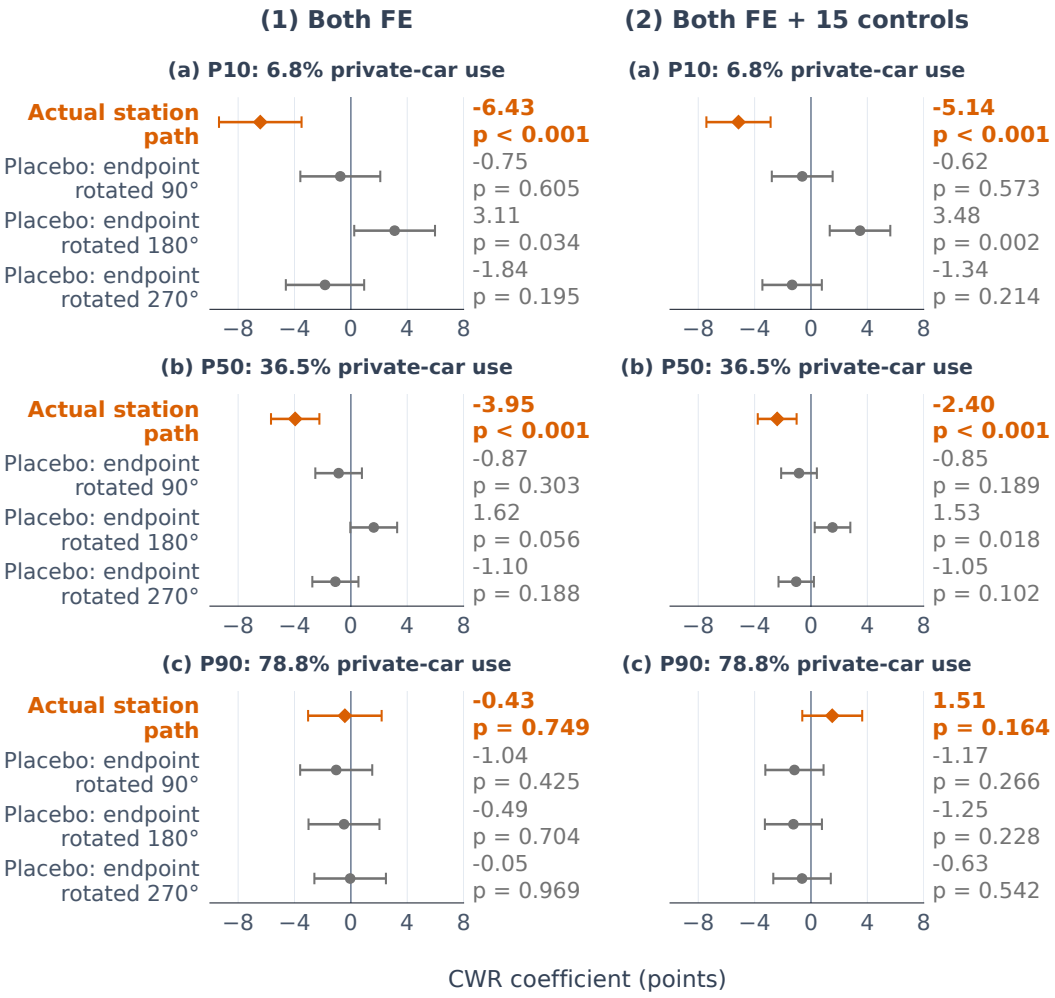


Figure B.3: Actual and placebo station-path coefficients by private-car use

Notes: Points show station-path CWR coefficients at the women-weighted 10th, 50th, and 90th percentiles of car use (6.8, 36.5, and 78.8 percent); bars are pointwise 95 percent confidence intervals. Both specifications include facility and parent-500m fixed effects and interact all five distance bands and four station paths with car-use rate. The left column has no controls; the right includes all 15. Both use 315,357 cells weighted by women ages 15–49, with two-way clustering by the fixed-effect units. Printed p-values test each coefficient against zero.

B.4 Joint heterogeneity by private-car use, female age, and education

Table B.3 extends Equation (B.1) to allow all five distance-band coefficients to vary jointly with private-car use, the cell-level mean age of women ages 15–49, and the local university-graduate share. All 15 interactions enter the same regression, together with the 15 baseline controls. The female age-share controls also account for mean age, the university-graduate share is included directly, and parent-500m fixed effects absorb car use.

Table B.3: Distance-band heterogeneity by private-car use, female age, and education

	Coefficients from a single regression			
	Distance at means	Distance × car use (10 pp)	Distance × female mean age (1 year)	Distance × graduate share (10 pp)
A. Distance-band main effects and interactions				
0–0.25km	−7.976**	2.475**	3.206***	4.476**
	(3.145)	(1.160)	(0.528)	(1.935)
0.25–0.50km	−4.804	1.634	2.547***	5.069***
	(3.081)	(1.140)	(0.506)	(1.905)
0.50–0.75km	−3.859	0.950	1.402***	3.844**
	(3.012)	(1.114)	(0.515)	(1.940)
0.75–1.00km	−2.814	0.585	0.058	2.421
	(2.783)	(1.060)	(0.536)	(1.974)
1.00–1.25km	0.313	−0.162	0.363	1.515
	(2.103)	(0.872)	(0.559)	(1.863)
B. Change in the 0–0.25km coefficient from P10 to P90				
P90 minus P10	—	17.881**	12.468***	13.150**
		(8.383)	(2.053)	(5.685)
Five interactions: joint p	—	0.0013	< 0.001	0.0331
Holm-adjusted joint p	—	0.0025	< 0.001	0.0331
Observations	331,850			
Fixed effects	Facility and parent-500m mesh			
Baseline controls	All 15, including log population density			

Notes: The outcome is the 2020 CWR (children ages 0–4 per 1,000 women ages 15–49). All four columns report one regression with five distance-band indicators and their 15 interactions with private-car use, female mean age, and the local university-graduate share. The reference band is 1.25–1.50km. The distance coefficients are evaluated at the women-weighted means: car use 41.086 percent, female mean age 33.876 years, and university-graduate share 25.331 percent. Interaction estimates and standard errors are expressed per 10 percentage points (pp) of car use or the university-graduate share, and per year of mean age.

All 15 baseline controls are included. The six female age-share controls also account for mean age, the university-graduate share is included directly, and parent-500m fixed effects absorb car use. Female mean age covers women ages 15–49 and uses five-year age-bin midpoints. The university-graduate share includes university and graduate-school graduates among school graduates ages 15 or older of both sexes; the denominator includes graduates whose final school is unreported. Private-car use is the parent-500m share of commuting workers and students using private cars, excluding people working at home.

Panel B varies one characteristic from its women-weighted 10th to 90th percentile, holding the other two at their means: 6.9–79.2 percent for car use, 32.0–35.9 years for mean age, and 11.8–41.1 percent for the university-graduate share. Estimation weights are the number of women ages 15–49, with two-way clustering by nearest facility and parent-500m mesh; standard errors are in parentheses. Each joint test covers the five interactions for one characteristic; Holm adjustment covers the three joint tests. Stars use unadjusted two-sided coefficient tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

B.5 Placebo analyses and 2015 cross-sectional replication

B.5.1 Future entry of disability facilities placebo

This placebo checks whether the 2015 CWR was already lower near candidate future disability facilities observed in 2021 using new entry of disability facilities described in Appendix A.7.

I estimate the fully controlled distance-band specification using the 2015 CWR and women ages 15–49 in 2015 as weights, including cells containing candidates. The model includes nearest-candidate and parent-500m fixed effects and all 14 available controls. A comparable 2015 university-graduate share is unavailable for the 2015 data. Table B.4 reports the estimates for the 198,147 cells shown in Panel A of Figure 6. None of the five coefficients is statistically significant at the 5 percent level.

Table B.4: Future Entry of Disability Facilities Placebo: 2015 CWR

Distance band (km)	Both FE + controls
0–0.25	−1.032
	(1.955)
0.25–0.50	1.034
	(1.725)
0.50–0.75	0.272
	(1.549)
0.75–1.00	0.047
	(1.352)
1.00–1.25	0.241
	(1.013)
1.25–1.50 (reference)	0
Facility fixed effects	Yes
Parent-500m fixed effects	Yes
Controls	Yes
Observations	198,147
Facilities with assigned cells	8,643

Notes: The outcome is the 2015 CWR; the reference band is 1.25–1.50km. Candidate 2021 facilities pass the 2011 and 2015 historical screens (Appendix A.7). Osaka is excluded. The model includes all 14 available controls. Weights are 2015 counts of women ages 15–49; inference uses two-way clustering by facility and parent-500m mesh (standard errors in parentheses). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Station-path placebo. I add the four station-path indicators to the fully controlled 2015 placebo specification, retaining the five facility-distance bands, nearest-candidate and parent-500m fixed effects, and all 14 controls. The paths connect the same candidate future facilities to their nearest stations, using the 2020 station locations and the January 1, 2021 walking network.

Table B.5 reports the four path coefficients from the joint model. The actual-path estimate is −0.254 (SE 0.913, $p = 0.781$), and the 90-, 180-, and 270-degree placebo-path estimates are 0.147, −0.025, and −1.118. None is statistically significant at the 5 percent level.

Table B.5: Future-Facility Station-Path Placebo: 2015 CWR

Variable	(1)
Actual station path	−0.254
	(0.913)
Placebo path: 90°	0.147
	(0.917)
Placebo path: 180°	−0.025
	(0.969)
Placebo path: 270°	−1.118
	(0.929)
Facility fixed effects	Yes
Parent-500m fixed effects	Yes
Controls	Yes
Observations	189,830
Facility groups	8,154

Notes: The outcome is the 2015 CWR around candidate future facilities. Four station paths and five distance bands (reference: 1.25–1.50km; coefficients omitted) enter jointly with all 14 controls. The sample requires all four station and four bus paths; Osaka is excluded. Paths use 2020 stations and the 2021 network. Weights are 2015 counts of women ages 15–49; inference uses two-way clustering by facility and parent-500m mesh (standard errors in parentheses). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

B.5.2 2020 CWR around the candidate future facilities

Table B.6 uses the same new-entry facilities as the 2015 placebo, but updates the outcome, mesh sample, population weights, and controls to the 2020 specification. The model includes both fixed effects and all 15 baseline controls. The fully controlled 0–250m coefficient is −3.742 (SE 1.952, $p = 0.055$), smaller in magnitude than the fully controlled distance-only estimate in column (2) of Table 2. The first three distance bands jointly reject zero ($p = 0.027$).

Table B.6: 2020 CWR around Candidate Future Disability Facilities

Distance band (km)	Both FE + controls
0–0.25	−3.742*
	(1.952)
0.25–0.50	−1.450
	(1.760)
0.50–0.75	−0.267
	(1.614)
0.75–1.00	0.034
	(1.404)
1.00–1.25	−0.354
	(1.048)
1.25–1.50 (reference)	0
Facility fixed effects	Yes
Parent-500m fixed effects	Yes
Controls	Yes
Observations	196,427
Facilities with assigned cells	8,618

Notes: The outcome is the 2020 CWR; the reference band is 1.25–1.50km. Candidate 2021 facilities pass the 2011 and 2015 historical screens (Appendix A.7). Osaka is excluded. The model includes all 15 controls. Weights are women ages 15–49; inference uses two-way clustering by facility and parent-500m mesh (standard errors in parentheses). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

B.5.3 2015 cross-sectional replication

Table B.7 reports the replication of Section 4 but using the 2015 data. I obtain results similar to those in the main 2020 analysis.

Table B.7: 2015 Cross-Sectional Replication: Distance to Disability Facilities

Distance band (km)	(1)	(2)
0–0.25	−9.218***	−8.744***
	(2.593)	(2.083)
0.25–0.50	−5.776**	−4.499**
	(2.413)	(1.939)
0.50–0.75	−2.853	−1.651
	(2.244)	(1.803)
0.75–1.00	−1.257	−1.095
	(1.979)	(1.607)
1.00–1.25	−1.797	−1.960
	(1.529)	(1.231)
1.25–1.50 (reference)	0	0
Facility fixed effects	Yes	Yes
Parent-500m fixed effects	Yes	Yes
Controls	No	Yes
Observations	265,690	265,690
Facilities with assigned cells	17,482	17,482

Notes: The outcome is the 2015 CWR; the reference band is 1.25–1.50km. Distances use 2015 facility records (Appendix A.7). Osaka is excluded. Both columns use the same sample. Column (1) has no controls; column (2) includes all 14 available controls. Weights are 2015 counts of women ages 15–49; inference uses two-way clustering by facility and parent-500m mesh (standard errors in parentheses). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

B.5.4 Elderly care facility placebo

Table B.8 repeats the analysis in Section 4 but using the elderly care facilities. Its fully controlled column (2) appears in Panel B of Figure 6. I do not find the negative coefficient around the elderly care facilities.

Table B.8: Distance to the Nearest Elderly Care Facility: Placebo Specifications

Variable	(1)	(2)
0–0.25	2.594	1.246
	(3.506)	(2.718)
0.25–0.50	−0.552	−1.442
	(3.083)	(2.386)
0.50–0.75	0.065	−0.271
	(2.632)	(2.027)
0.75–1.00	0.509	−0.301
	(2.192)	(1.676)
1.00–1.25	−0.024	−0.547
	(1.562)	(1.241)
1.25–1.50 (reference)	0	0
Facility fixed effects	Yes	Yes
Parent-500m fixed effects	Yes	Yes
Controls	No	Yes
Observations	176,077	176,077

Notes: The outcome is the 2020 CWR; the reference band is 1.25–1.50km. Elderly facilities lie more than 1km from every eligible disability service facility. Both columns use the same sample. Column (1) has no controls; column (2) includes all 15. Models use women ages 15–49 as weights and two-way clustering by facility and parent-500m mesh (standard errors in parentheses). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Actual and placebo station paths. Table B.9 reports the path-analysis results for elderly care facilities. The actual-path coefficient is close to zero (−0.376, SE 1.248, $p = 0.763$), so this placebo does not reproduce the negative station-path association around disability facilities.

Table B.9: Elderly-Care Facilities: Actual and Placebo Station Paths

Variable	(1)
Actual station path	−0.376
	(1.248)
Placebo path: 90°	1.830
	(1.333)
Placebo path: 180°	−0.157
	(1.467)
Placebo path: 270°	1.902
	(1.331)
Facility fixed effects	Yes
Parent-500m fixed effects	Yes
Controls	Yes
Observations	163,197
Facility groups	10,648

Notes: The outcome is the 2020 CWR. Four station paths and five distance bands (reference: 1.25–1.50km; coefficients omitted) enter jointly with all 15 controls. Elderly facilities lie more than 1km from every eligible disability service facility. The sample requires all four station and four bus paths. Weights are women ages 15–49; inference uses two-way clustering by facility and parent-500m mesh (standard errors in parentheses). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

B.6 Residential sorting

Table B.10 holds fixed a sample of 240,193 cells with positive 2015 denominators for both child cohorts, all four sorting measures observed, complete baseline variables, and support for both fixed effects. Appendix A.8 defines the measures.

The table compares CWR regressions with the net child migration rate for ages 5–9 in 2020 alone, both net child migration rates, and both rates plus female population change and the female recent-mover share. The specification with both rates also appears in Panel A of Table 3.

Table B.10: CWR Sensitivity to Residential Sorting Controls

Model	(1)	(2)
No migration controls	−8.067**	−8.421***
	(3.309)	(2.534)
Add net child migration rate (5–9 as of 2020)	−6.766**	−7.841***
	(3.279)	(2.537)
Add both net child migration rates	−6.771**	−7.861***
	(3.279)	(2.539)
Add both net child migration rates, female population change, and recent-mover share	−9.934***	−7.380***
	(2.922)	(2.403)
Nearest-facility fixed effects	Yes	Yes
Parent-500m mesh fixed effects	Yes	Yes
Controls	No	Yes
Observations	240,193	240,193

Notes: Entries compare 2020 CWR at 0–250m with 1.25–1.50km. All rows use the same sample with positive 2015 child-cohort counts and all four sorting measures. The listed migration controls enter both columns. Column (1) has no baseline controls; column (2) includes all 15. Weights are women ages 15–49; standard errors in parentheses are clustered by facility and parent-500m mesh. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

B.7 Residential land prices and neighborhood valuation

Table B.11 uses predicted log land price, spatially interpolated from assessment prices, as the outcome on the main sample. Both columns retain the female-population weights and both fixed effects. Column (1) has no baseline controls and column (2) includes all 15.

Table B.11: Disability-Facility Proximity and Residential Land Prices

	(1) Both FE	(2) Both FE + controls
Dependent variable: log 2020 residential land price		
0–250m	0.0432***	0.0329***
	(0.0038)	(0.0037)
250–500m	0.0358***	0.0276***
	(0.0037)	(0.0036)
500–750m	0.0262***	0.0206***
	(0.0034)	(0.0033)
750–1,000m	0.0151***	0.0116***
	(0.0031)	(0.0030)
1,000–1,250m	0.0068***	0.0052**
	(0.0023)	(0.0022)
Facility and parent-500m fixed effects	Yes	Yes
Controls	No	Yes
Observations	331,850	331,850

Notes: The outcome is predicted log residential land price in 2020, interpolated from assessment prices. The reference band is 1.25–1.50km. Column (1) has no controls; column (2) includes all 15. Both use the main sample and women ages 15–49 weights. Standard errors in parentheses are clustered by facility and parent-500m mesh. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

B.8 Recorded crime as an alternative mechanism

B.8.1 Nationwide recorded theft

Table B.12 uses the 2020 rate for all seven theft methods as the outcome, with population weights and the two supplementary specifications. The common sample contains 237,697 cells with at least 50 residents, after recursively removing singleton groups for facility and parent-500m fixed effects. Panel C of Table 3 adds this theft rate to the CWR regressions. Comparisons with and without theft use the same cells and women ages 15–49 weights.

Table B.12: Nationwide Theft

Distance band (km)	(1)	(2)
0–0.25	−0.154	−0.092
	(0.156)	(0.157)
0.25–0.50	−0.224	−0.177
	(0.151)	(0.152)
0.50–0.75	−0.246*	−0.211
	(0.141)	(0.141)
0.75–1.00	−0.237*	−0.217
	(0.142)	(0.142)
1.00–1.25	−0.073	−0.073
	(0.091)	(0.090)
1.25–1.50 (reference)	0	0
Nearest-facility fixed effects	Yes	Yes
Parent-500m fixed effects	Yes	Yes
Controls	No	Yes
Observations	237,697	237,697

Notes: The outcome is recorded theft per 1,000 residents in 2020, covering seven theft categories. The reference band is 1.25–1.50km. Both columns use a common sample of cells with at least 50 residents and population weights. Column (1) has no controls; column (2) includes all 15. Standard errors in parentheses are clustered by facility and parent-500m mesh. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

B.8.2 Tokyo town/chome crime

The Tokyo analysis uses one observation per town/chome and outcomes in incidents per 1,000 residents; source matching is described in Appendix A.10. Eligible areas have at least 50 residents and a representative point less than 1.5km from an admissible disability service facility. The six available baseline controls are log population density, log area, the three zoning indicators, and log station-access distance. Both specifications use population weights and include nearest-facility and parent-1km mesh fixed effects, with two-way clustering by those units. Specification (1) has no baseline controls; specification (2) includes all six. Table B.13 reports the five-band profile for each outcome under both specifications, showing no evidence that crime rates are higher near the disability service facilities.

Table B.13: Tokyo Town/Chome Crime Outcomes, 2020

Distance band	All Penal Code offenses	Violent or serious offenses	Burglary	Non-burglary theft
(1) Facility and parent-1km FE; no controls				
0–250m	−1.060	−0.945**	−0.616**	0.368
	(3.814)	(0.440)	(0.311)	(2.673)
250–500m	−0.787	−0.874**	−0.648**	0.528
	(3.795)	(0.435)	(0.311)	(2.662)
500–750m	−1.767	−1.027**	−0.727**	0.079
	(3.777)	(0.429)	(0.310)	(2.648)
750–1,000m	0.655	−0.840**	−0.784***	1.954
	(3.367)	(0.402)	(0.299)	(2.350)
1,000–1,250m	−0.233	−0.896**	−0.702*	1.881
	(3.938)	(0.404)	(0.410)	(2.733)
1,250–1,500m (reference)	0	0	0	0
Facility fixed effects	Yes	Yes	Yes	Yes
Parent-1km mesh fixed effects	Yes	Yes	Yes	Yes
Controls	No	No	No	No
Observations	2,408	2,408	2,408	2,408
(2) Facility and parent-1km FE; controls				
0–250m	−0.308	−0.876**	−0.658***	0.947
	(3.692)	(0.426)	(0.237)	(2.618)
250–500m	−0.176	−0.805*	−0.705***	1.027
	(3.680)	(0.421)	(0.235)	(2.613)
500–750m	−1.766	−1.011**	−0.817***	0.170
	(3.684)	(0.422)	(0.235)	(2.615)
750–1,000m	0.716	−0.833**	−0.869***	2.067
	(3.286)	(0.391)	(0.228)	(2.349)
1,000–1,250m	−2.232	−1.092***	−0.858***	0.566
	(3.909)	(0.392)	(0.320)	(2.782)
1,250–1,500m (reference)	0	0	0	0
Facility fixed effects	Yes	Yes	Yes	Yes
Parent-1km mesh fixed effects	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Observations	2,408	2,408	2,408	2,408

Notes: Outcomes are recorded incidents per 1,000 residents in 2020. Both specifications use the same town/chomes with at least 50 residents and population weights. The 1.25–1.50km reference band contains only five town/chomes. Specification (1) has no controls; specification (2) includes all six available controls. Violent or serious offenses combine the two official categories. Standard errors in parentheses are clustered by facility and parent-1km mesh. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

B.9 Sensitivity to unobserved confounding

To assess the sensitivity of my results to unobserved confounders, Table B.14 applies the method of Oster (2019) to both distance-only 0–250m CWR specifications, corresponding to columns (1)–(2) of Table 2. Let D be the nearest-band indicator. In Panel A, the always-included regressors M comprise facility and parent-500m fixed effects and the four other distance indicators. In Panel B, M also includes all 15 baseline controls. Each exercise adds one benchmark G to (D, M): both net child migration rates, all four sorting measures, predicted log residential land price, recorded theft, or a joint block of all four sorting measures, land price, and theft. Overall, I find that the negative impact on the CWR is robust to considering the presence of unobserved confounders.

Calculation. We apply Oster's method after applying the Frisch–Waugh–Lovell theorem. After weighted residualization of CWR, D, and G on the panel-specific M, let R_s^2 and R_f^2 be the conditional R^2 before and after adding G. In each panel, the unobserved component is assumed to add the same explanatory power as the observed benchmark:

$$R_{\max}^2 = R_f^2 + (R_f^2 - R_s^2). \tag{B.3}$$

The parameter δ measures selection on the unobserved outcome component relative to the observed benchmark. Each panel solves the exact equation at $\delta = 1$, selecting the closest valid root in the direction of the observed coefficient change and checking the proportional-selection sign condition. The table also reports the δ for which a zero coefficient solves the exact equation. With multiple roots, this value need not mark a sign reversal (Masten and Poirier, 2026).

Table B.14: Oster Sensitivity of the CWR Coefficient to Unobserved Confounding

Quantity	Residential sorting		Land price	Recorded theft	Joint controls
	Both child cohorts	All four measures			All six measures
Panel A: Facility and parent-500m FE; no controls					
Observations	240,193	240,193	331,850	237,697	213,183
CWR coefficient without benchmark G	−8.0670	−8.0670	−6.7106	−8.1295	−8.5446
CWR coefficient with benchmark G	−6.7706	−9.9344	−6.7104	−8.1308	−9.3259
Conditional R^2_s (without G)	0.00005389	0.00005389	0.00003292	0.00005431	0.00006261
Conditional R^2_f (with G)	0.01845071	0.15513069	0.00003292	0.00005461	0.16320541
Conditional R^2 gain	0.01839681	0.15507680	7.86×10^{-12}	0.00000030	0.16314280
Conditional $R^2_{\max}$	0.03684752	0.31020749	0.00003292	0.00005492	0.32634822
Adjusted coefficient at $\delta = 1$	−5.474	−11.803	−6.710	−8.132	−10.111
δ yielding a zero coefficient	5.21	−5.32	1.01	−32.68	−12.25
Panel B: Both fixed effects and controls					
Observations	240,193	240,193	331,850	237,697	213,183
CWR coefficient without benchmark G	−8.4213	−8.4213	−7.7571	−8.1783	−8.4333
CWR coefficient with benchmark G	−7.8607	−7.3800	−7.3296	−8.1736	−6.7421
Conditional R^2_s (without G)	0.00008780	0.00008780	0.00006266	0.00008361	0.00009332
Conditional R^2_f (with G)	0.00779948	0.08044899	0.00019833	0.00009567	0.08411140
Conditional R^2 gain	0.00771168	0.08036119	0.00013567	0.00001206	0.08401808
Conditional $R^2_{\max}$	0.01551116	0.16081018	0.00033399	0.00010773	0.16812948
Adjusted coefficient at $\delta = 1$	−7.300	−6.338	−6.902	−8.169	−5.047
δ yielding a zero coefficient	13.89	7.07	12.44	224.00	3.96

Notes: Entries concern the 0–250m CWR coefficient relative to 1.25–1.50km. Panel A has no controls; Panel B includes all 15. Samples and women ages 15–49 weights are fixed within each benchmark comparison. Unobserved factors are assumed to add the same conditional R^2 gain as the benchmark. Adjusted coefficients are point estimates; the zero-coefficient δ need not mark a sign reversal. Appendix B.9 explains the calculation and root selection.

B.10 Number of nearby facilities

This analysis replaces the five distance-band indicators in Equation (2) with indicators for one, two, three, and four or more disability service facilities within 250m of the cell centroid. The reference category contains no facility within 250m. Table B.15 reports both specifications using the main sample and the same nearest-facility and parent-500m fixed effects.

Table B.15: CWR and the number of disability service facilities within 250m

Facilities within 250m	Cells	(1) No controls	(2) 15 controls
0 (reference)	262,744	0	0
1	43,465	−3.296***	−2.867***
		(0.824)	(0.623)
2	14,714	−3.829***	−4.339***
		(1.247)	(0.966)
3	5,969	−5.014***	−5.842***
		(1.821)	(1.451)
4 or more	4,958	−8.391***	−8.420***
		(2.365)	(1.830)
Facility fixed effects		Yes	Yes
Parent-500m fixed effects		Yes	Yes
Observations		331,850	331,850
Equal coefficients: 1 and 2 (p)		0.681	0.145
Equal coefficients: 2 and 4+ (p)		0.064	0.032
Equal coefficients: 2, 3, and 4+ (p)		0.180	0.096
Equal coefficients: 1, 2, 3, and 4+ (p)		0.193	0.011

Notes: The outcome is the 2020 CWR, measured as children ages 0–4 per 1,000 women ages 15–49. Counts cover the disability service facilities used in the main analysis within a great-circle distance of at most 250m from each cell centroid. Separate facility records at the same coordinates are counted separately. The four count indicators replace the distance-band indicators; zero facilities is the reference, and counts of four or more are pooled. Both columns use the main sample, weighted by women ages 15–49, with two-way clustering by nearest facility and parent-500m mesh (standard errors in parentheses). Column (2) includes all 15 baseline controls. Equality tests use the full clustered covariance matrix. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

B.11 Robustness checks

Table B.16 excludes facility-containing cells and rebuilds the common fixed-effect support.

Table B.16: Distance to the Nearest Disability Facility with Facility-Containing Cells Excluded

Distance band (km)	(1)	(2)
0–0.25	−5.706*	−7.001***
	(3.118)	(2.433)
0.25–0.50	−3.036	−4.138*
	(2.980)	(2.314)
0.50–0.75	−2.820	−3.510
	(2.862)	(2.219)
0.75–1.00	−1.792	−2.707
	(2.573)	(2.001)
1.00–1.25	−0.145	−0.488
	(1.955)	(1.556)
1.25–1.50 (reference)	0	0
Facility fixed effects	Yes	Yes
Parent-500m fixed effects	Yes	Yes
Controls	No	Yes
Observations	294,946	294,946

Notes: Entries are 2020 CWR differences relative to 1.25–1.50km. The common sample excludes cells containing an eligible disability service facility. Column (1) has no controls; column (2) includes all 15. Both use women ages 15–49 as weights. Standard errors in parentheses are clustered by facility and parent-500m mesh. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table B.17 compares the standard CWR with the alternative age-20–39 denominator defined in Appendix A.11.

Table B.17: Alternative CWR Denominator on Common Support

Outcome	(1)	(2)	*N*
CWR using women ages 15–49	−7.169** (3.090)	−7.677*** (2.402)	315,550
CWR using women ages 20–39	−18.582*** (5.507)	−13.140*** (4.609)	315,550
Facility fixed effects	Yes	Yes	
Parent-500m fixed effects	Yes	Yes	
Controls	No	Yes	

Notes: Entries compare 2020 CWR at 0–250m with 1.25–1.50km. CWR is children ages 0–4 per 1,000 women in the stated age group. Both rows use the same cells with positive counts of women ages 20–39; each outcome uses its own denominator as weights. Column (1) has no controls; column (2) includes all 15. Standard errors in parentheses are clustered by facility and parent-500m mesh. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table B.18 extends the estimation window to 2km and adds two outer-band indicators while retaining the main 1.25–1.50km reference band.

Table B.18: Extending the Estimation Window to 2km with an Unchanged Reference Band

Distance band (km)	(1)	(2)
0–0.25	−6.656**	−7.714***
	(3.035)	(2.366)
0.25–0.50	−2.885	−4.052*
	(2.948)	(2.289)
0.50–0.75	−2.788	−3.555
	(2.828)	(2.195)
0.75–1.00	−1.745	−2.647
	(2.541)	(1.978)
1.00–1.25	−0.185	−0.452
	(1.922)	(1.529)
1.25–1.50 (reference)	0	0
1.50–1.75	0.682	1.121
	(2.161)	(1.789)
1.75–2.00	−2.439	−1.412
	(3.072)	(2.564)
Facility fixed effects	Yes	Yes
Parent-500m fixed effects	Yes	Yes
Controls	No	Yes
Observations	366,528	366,528

Notes: Entries are 2020 CWR differences. The estimation window extends to 2km, retaining 1.25–1.50km as the reference and adding two outer-band indicators. Column (1) has no controls; column (2) includes all 15. Both use the same sample and women ages 15–49 weights. Standard errors in parentheses are clustered by facility and parent-500m mesh. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table B.19 reports spatial inference for both distance-only specifications: Panel A has no baseline controls and Panel B includes all 15. Conley-type standard errors allow residual spatial correlation across facility and parent-mesh boundaries, using the three cutoffs in Section 5. The coefficients are unchanged, and the nearest-band coefficient remains negative and statistically significant at all three cutoffs.

Table B.19: Spatial-Inference Robustness

Inference	Coefficient	S.E.	p-value	N
Panel A: Facility and parent-500m FE; no controls				
Facility and parent-500m clusters	-6.711^{**}	3.062	0.028	331,850
Conley, 0.5km	-6.711^{**}	2.767	0.015	331,850
Conley, 1km	-6.711^{**}	2.972	0.024	331,850
Conley, 2km	-6.711^{**}	3.099	0.030	331,850
Panel B: Both fixed effects and controls				
Facility and parent-500m clusters	-7.757^{***}	2.380	0.001	331,850
Conley, 0.5km	-7.757^{***}	2.142	< 0.001	331,850
Conley, 1km	-7.757^{***}	2.288	< 0.001	331,850
Conley, 2km	-7.757^{***}	2.364	0.001	331,850

Notes: Entries are 0–250m coefficients for 2020 CWR relative to 1.25–1.50km. Both panels use the main sample and women ages 15–49 weights. Panel A has no controls; Panel B includes all 15. Within each panel, coefficients are unchanged; only inference differs. The first row uses two-way clustering by facility and parent-500m mesh; the others use Bartlett-kernel Conley standard errors with 0.5, 1, and 2km cutoffs. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

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